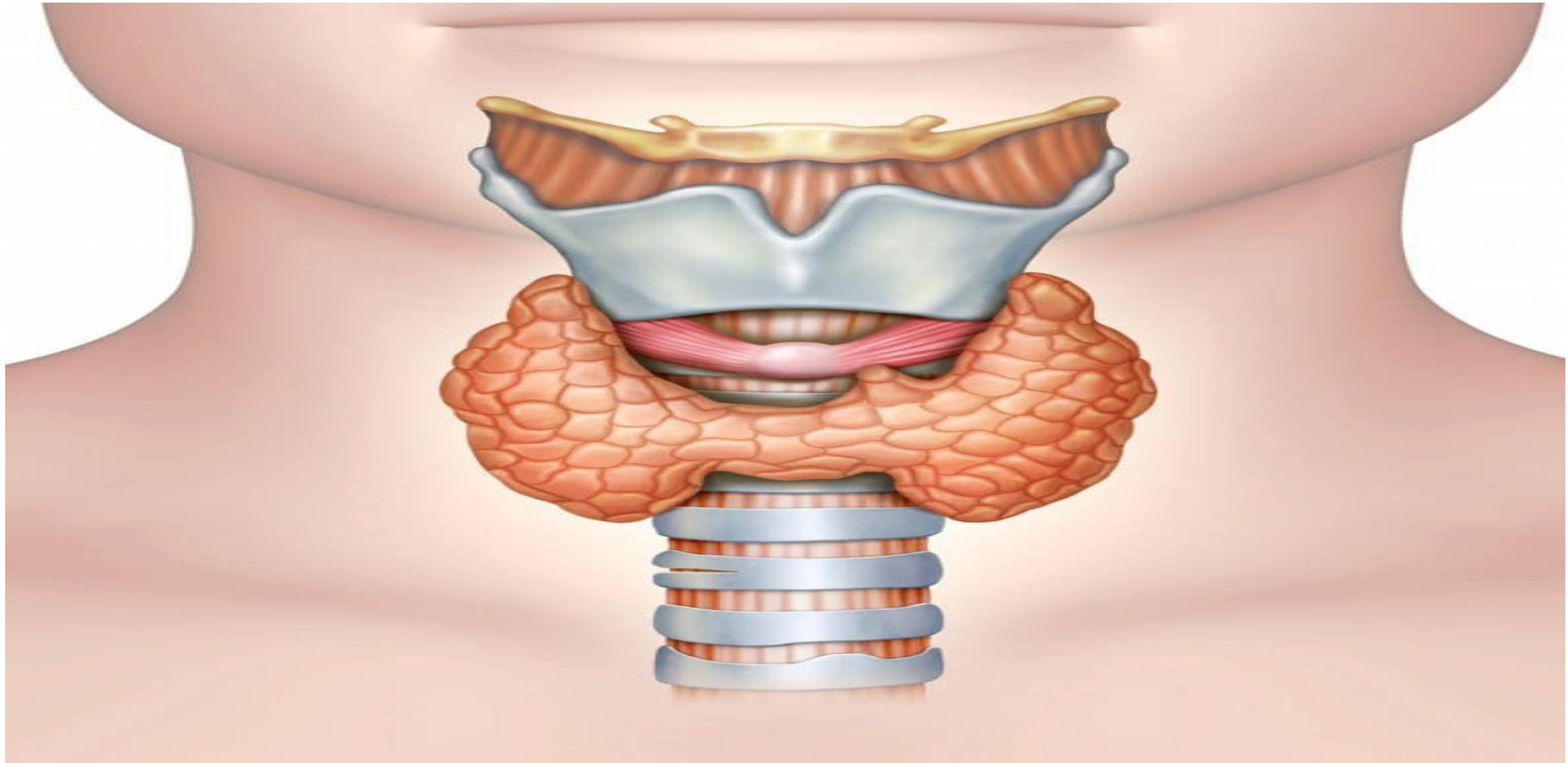


Thyroid Gland and Hormones



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Learning Objectives

Describe thyroid hormones

Describe Thyroid Hormones Synthesis

Disorders of Thyroid Hormones

THYROID GLAND

INTRODUCTION

Thyroid is an endocrine gland situated at the root of the neck on either side of the trachea.

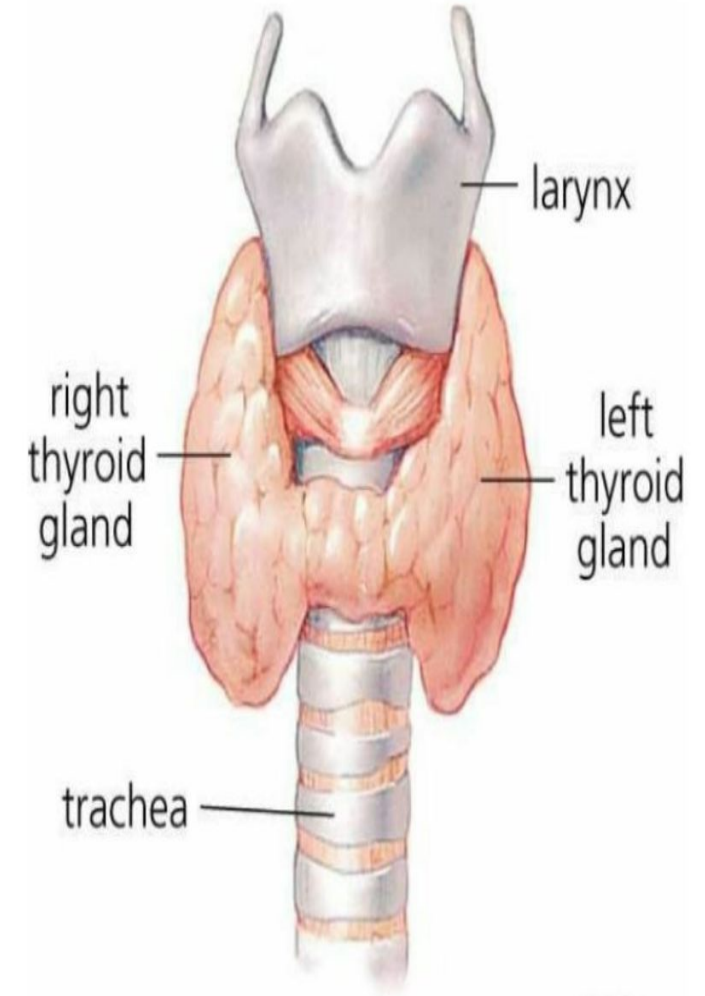
It has two lobes, which are connected in the middle by an isthmus.

It weighs about 20 to 40 g in adults.

Thyroid is larger in females than in males.

The structure and the function of the thyroid gland change in different stages of the sexual cycle in females.

Its function increases slightly during pregnancy and lactation and decreases during menopause.



HISTOLOGY OF THYROID GLAND

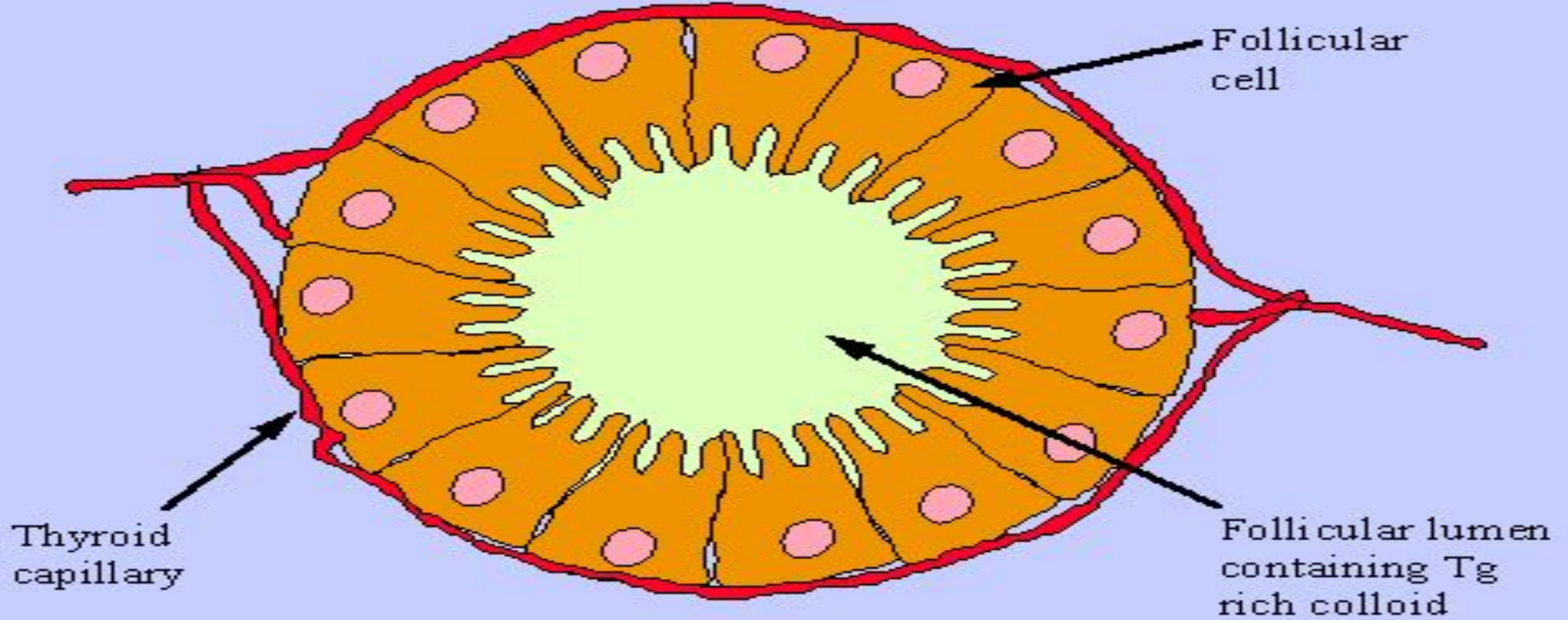
Thyroid gland is composed of large number of closed follicles. These follicles are lined with cuboidal epithelial cells, which are called the follicular cells.

Follicular cavity is filled with a colloidal substance known as thyroglobulin, which is secreted by the follicular cells.

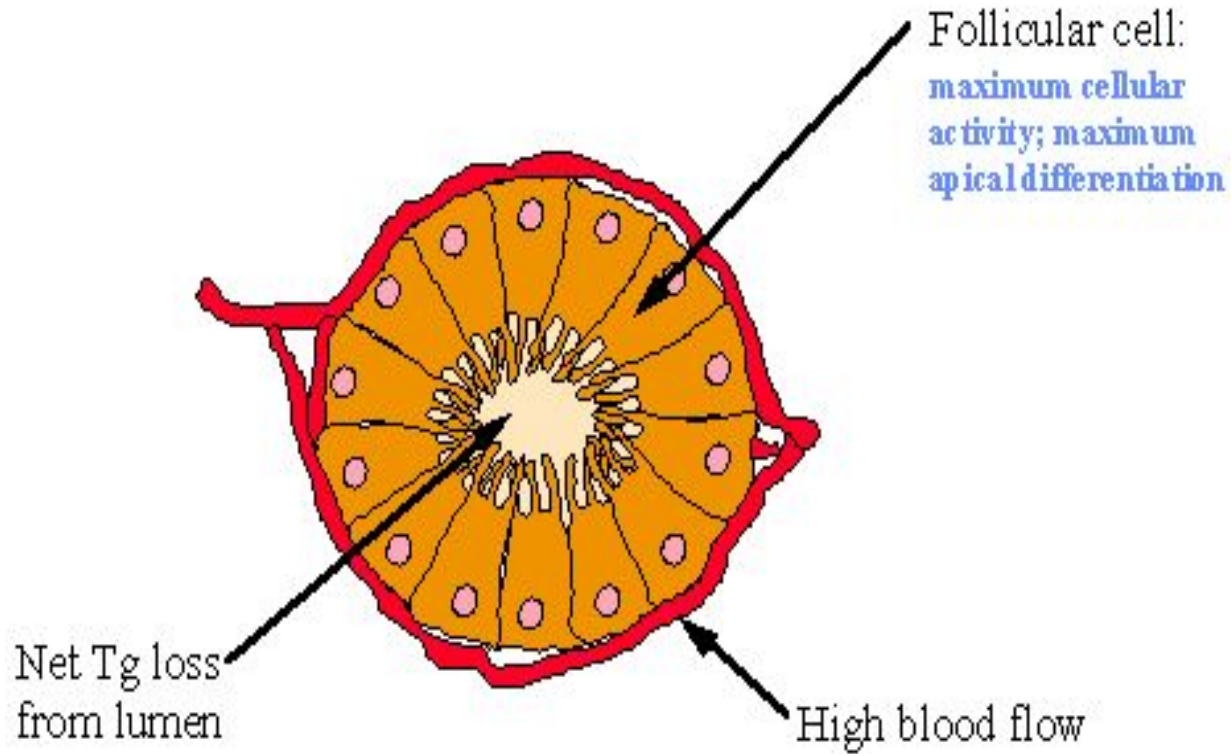
Follicular cells also secrete tetraiodothyronine (T4 or thyroxine) and triiodothyronine (T3).

In between the follicles, the parafollicular cells are present. These cells secrete calcitonin

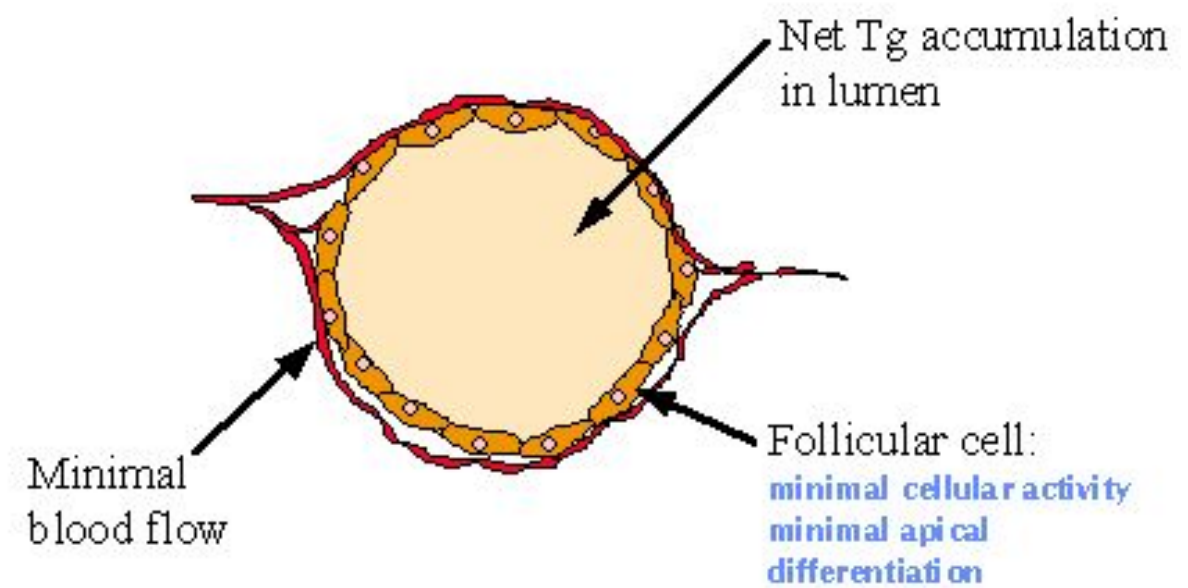
The functional unit of thyroid is thyroid follicle.
Normal follicle



Thyroid follicle with high TSH stimulation



Thyroid follicle without TSH

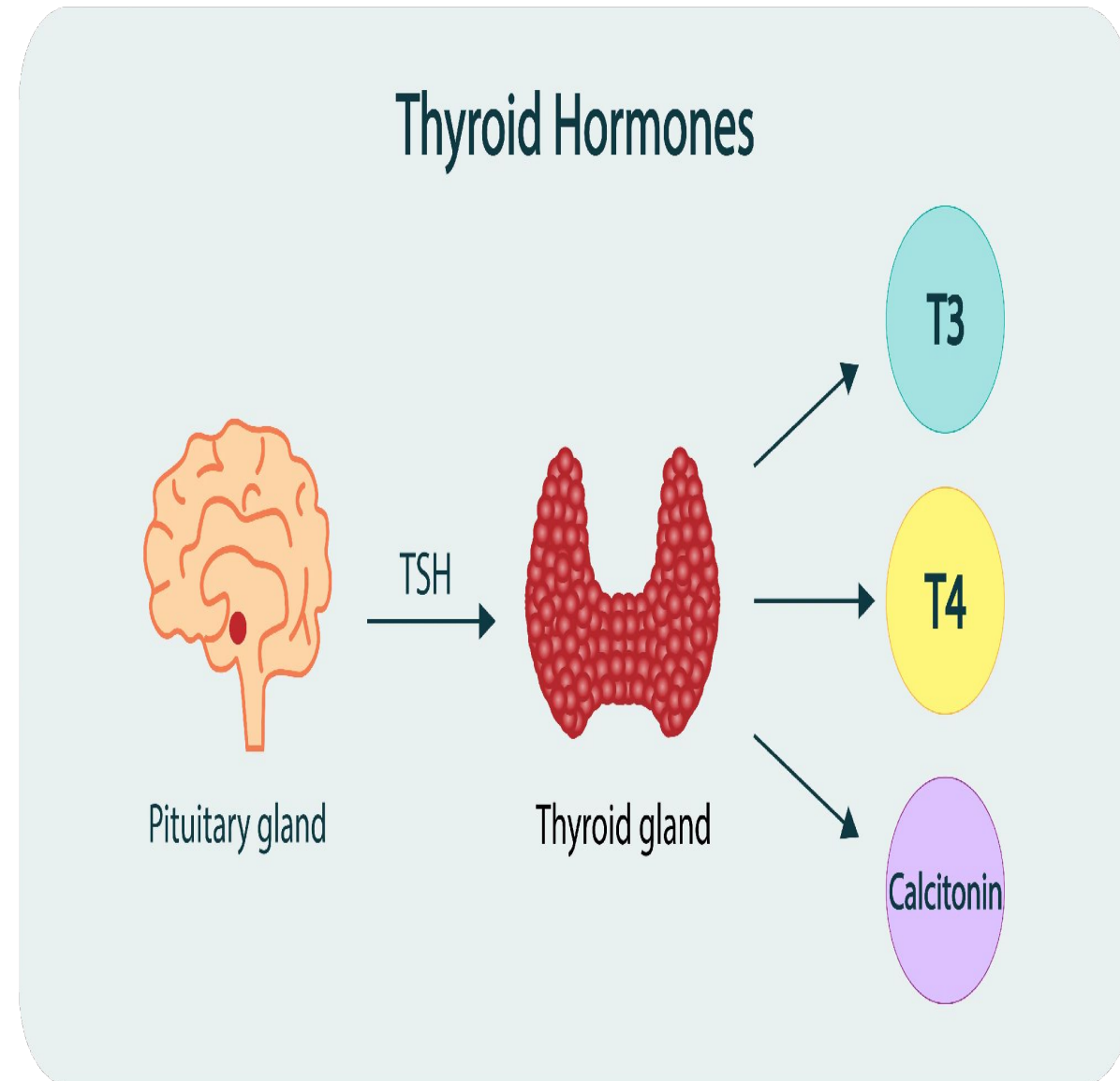


HORMONES OF THYROID GLAND

Thyroid gland secretes three hormones:

1. Tetraiodothyronine or T4 (thyroxine)
2. Tri-iodothyronine or T3
3. Calcitonin.

T4 is otherwise known as thyroxine and it forms about 90% of the total secretion, whereas T3 is only 9% to 10%.



SYNTHESIS OF THYROID HORMONES

- Synthesis of thyroid hormones takes place in thyroglobulin, present in follicular cavity.
- Iodine and tyrosine are essential for the formation of thyroid hormones.
- Iodine is consumed through diet. It is converted into iodide and absorbed from GI tract.
- Tyrosine is also consumed through diet and is absorbed from the GI tract.
- For the synthesis of normal quantities of thyroid hormones, approximately 1 mg of iodine is required per week or about 50 mg per year.
- To prevent iodine deficiency, common table salt is iodized with one part of sodium iodide to every 100,000 parts of sodium chloride

STAGES OF SYNTHESIS OF THYROID HORMONES

Synthesis of thyroid hormones occurs in following stages:

1. Thyroglobulin synthesis
2. Iodide trapping
3. Oxidation of iodide
4. Transport of iodine into follicular cavity
5. Iodination of tyrosine
6. Coupling reactions.
- 7-Secretion of thyroid hormones

Thyroid Hormone Synthesis

❖ Steps of Thyroid Hormone Synthesis

➤ The thyroid hormone synthesis involves following steps:

1. Iodide trapping
2. Conversion of iodide into iodine
3. Thyroglobulin synthesis
4. The coupling reaction
5. Proteolysis of thyroglobulin
6. Secretion of thyroid hormones

TRANSPORT OF THYROID HORMONES IN THE BLOOD

Thyroid hormones are transported in the blood by three types of proteins:

1. Thyroxine-binding globulin (TBG)
2. Thyroxine-bin
3. Albumin.

Factors decreasing the secretion of thyroid hormones:

1. Excess iodide intake
2. Stress
3. Somatostatin
4. Glucocorticoids
5. Dopamine.

These factors decrease the secretion of thyroid hormones, by inhibiting the release of TRH

REGULATION OF SECRETION OF THYROID HORMONES

Secretion of thyroid hormones is controlled by anterior pituitary and hypothalamus through feedback mechanism. Many factors are involved in the regulation of thyroid secretion.

ROLE OF PITUITARY GLAND

Thyroid-stimulating hormone (TSH) secreted by anterior pituitary is the major factor regulating the synthesis and release of thyroid hormones

ROLE OF HYPOTHALAMUS

Hypothalamus regulates thyroid secretion by controlling TSH secretion through thyrotropin-releasing hormone (TRH). From hypothalamus, TRH is transported through the hypothalamo-hypophyseal portal vessels to the anterior pituitary. After reaching the pituitary gland, the TRH causes the release of TSH.

FEEDBACK

Thyroid hormones regulate their own secretion through negative feedback control, by inhibiting the release of TRH from hypothalamus and TSH from anterior pituitary

ROLE OF IODIDE

Iodide is an important factor regulating the synthesis of thyroid hormones. When the dietary level of iodine is moderate, the blood level of thyroid hormones is normal. However, when iodine intake is high, the enzymes necessary for synthesis of thyroid hormones are inhibited by iodide itself, resulting in suppression of hormone synthesis. This effect of iodide is called Wolff Checkoff effect.

ROLE OF OTHER FACTORS

Many other factors are involved in the regulation of thyroid secretion in accordance to the needs of the body

Factors increasing the secretion of thyroid hormones:

1. Low basal metabolic rate
2. Leptin (Leptin is a hormone your body releases that helps it maintain your normal weight on a long-term basis).
3. α -melanocyte-stimulating hormone

Leptin (from adipose tissue) and α -melanocyte

stimulating hormone (from pituitary) increase the release of TRH and synthesis of T4

The low body temperature also stimulates the synthesis of thyroid hormones. However, this occurs only in infants.

Factors decreasing the secretion of thyroid hormones:

1. Excess iodide intake
2. Stress
3. Somatostatin
4. Glucocorticoids
5. Dopamine.

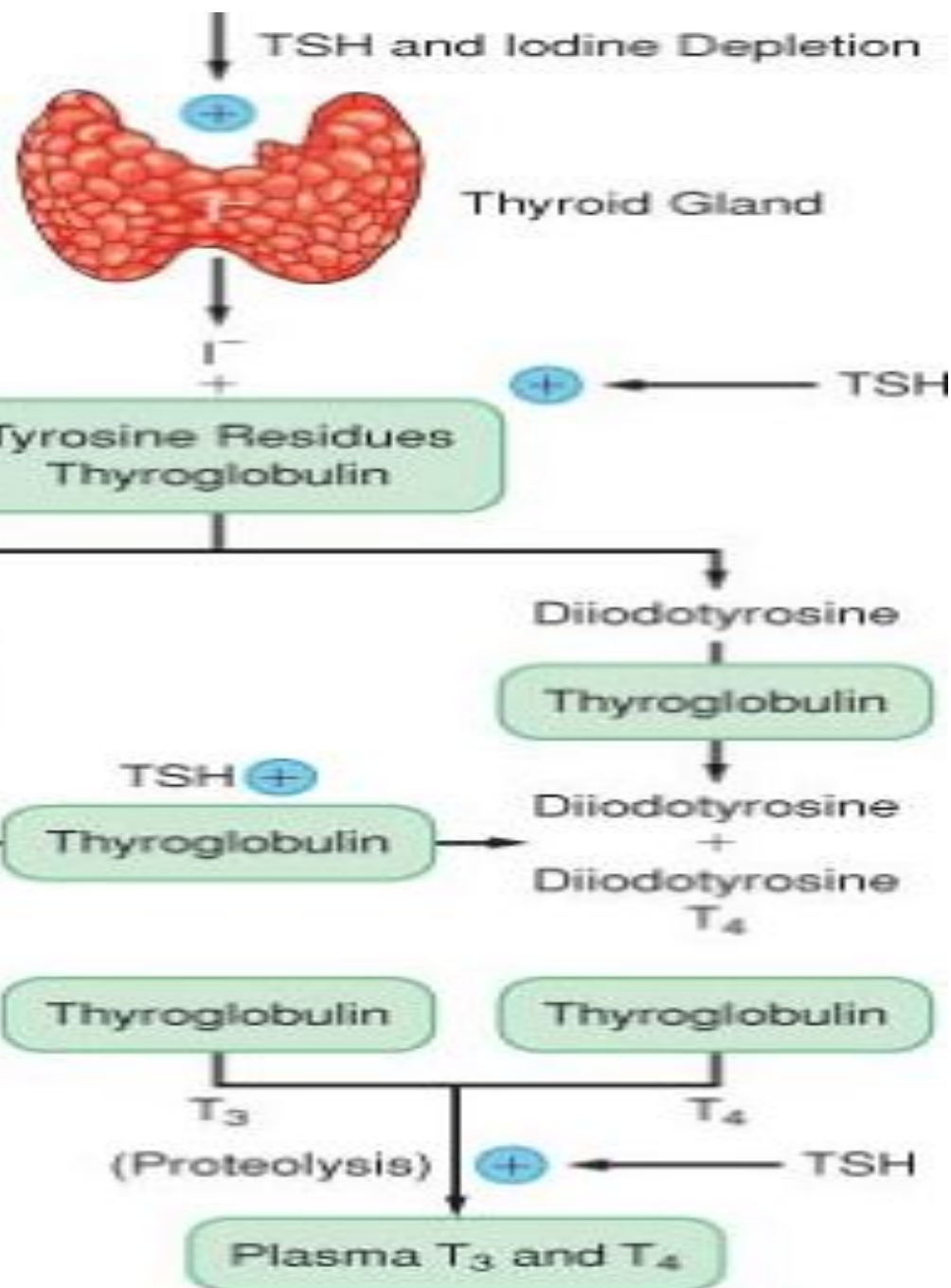
These factors decrease the secretion of thyroid hormones, by inhibiting the release of TRH.

1. Iodide trapping in Thyroid Gland

2. Oxidation of Iodide Binding to Tyrosine

3. Coupling (Intracellular Oxidation)

4. Release and Recycling



Thyroid Hormone Storage

Storage Location: Thyroid hormones are stored in the thyroid gland bound to a large glycoprotein called **thyroglobulin** inside the colloid-filled lumen of thyroid follicles.

Capacity: The thyroid gland is unique because it stores a large quantity of hormone—enough to supply the body's normal requirements for **2 to 3 months** (and up to several months). In combination with thyroglobulin, the thyroid hormones can be stored for several months. Thyroid gland is unique in this, as it is the only endocrine gland that can store its hormones for a long period of about 4 months.

So, when the synthesis of thyroid hormone stops, the signs and symptoms of deficiency do not appear for about 4 months.

Release Mechanism: When stimulated by thyroid-stimulating hormone (TSH), the colloid is reabsorbed into follicular cells via pinocytosis/endocytosis, where lysosomes digest thyroglobulin to release free T3 and T4 into the blood.

RELEASE OF THYROID HORMONES FROM THE THYROID GLAND

Thyroid hormones (T3 and T4) are released from the thyroid gland when thyroid-stimulating hormone (TSH) from the pituitary gland triggers follicular cells to reabsorb and break down stored thyroglobulin

Steps of Thyroid Hormone Release

TSH Stimulation: TSH binds to specific receptors on the surface of thyroid follicular cells (thyrocytes). This signal tells the cells to start the release process.

Endocytosis (Uptake): Follicle cells in the thyroid take up stored thyroglobulin (the protein scaffold containing inactive thyroid hormones) from the follicles. The thyrocytes use endocytosis (pinocytosis) to extend little borders and pull iodinated thyroglobulin back into the cell from the follicular lumen (colloid).

Lysosomal Fusion: Lysosomes break down the thyroglobulin structure, separating the active T3 and T4 hormones.

Proteolysis (Cleavage): Proteolytic enzymes in these joined sacs break down (hydrolyze) the large thyroglobulin protein scaffold. This action cleaves free triiodothyronine (T3) and thyroxine (T4), along with uncoupled MIT and DIT.

Secretion into Bloodstream: The free T3 and T4 hormones cross the basal membrane and pass through transporter proteins (like MCT8) into the surrounding fenestrated capillaries and bloodstream. About 80% of what is released is T4, and 20% is T3.

Transport: Once in circulation, most of these hormones bind to carrier proteins like thyroid-binding globulin to travel through the body. [\[1\]](#)

Recycling: Extra MIT and DIT molecules left behind are stripped of their iodine by deiodinase enzymes so the iodine and amino acids can be reused by the gland.

FUNCTIONS OF THYROID HORMONES

Thyroid hormones have two major effects on the body:

- I. To increase basal metabolic rate
- II. To stimulate growth in children.

The actions of thyroid hormones are:

1. ACTION ON BASAL METABOLIC RATE (BMR)

Thyroxine increases the metabolic activities in most of the body tissues, except brain, retina, spleen, testes and lungs. It increases BMR by increasing the oxygen consumption of the tissues. The action that increases the BMR is called **calorigenic action**.

2. ACTION ON PROTEIN METABOLISM

Thyroid hormone increases the synthesis of proteins in the cells.

3. ACTION ON CARBOHYDRATE METABOLISM

Thyroxine stimulates almost all processes involved in the metabolism of carbohydrate.

4. ACTION ON FAT METABOLISM

Thyroxine decreases the fat storage by mobilizing it from adipose tissues and fat depots. The mobilized fat is converted into free fatty acid and transported by blood. Thus, thyroxine increases the free fatty acid level in blood.

5. ACTION ON PLASMA AND LIVER FATS

Even though there is an increase in the blood level of free fatty acids, thyroxine specifically decreases the cholesterol, phospholipids and triglyceride levels in plasma.

In hypo secretion of thyroxine, the cholesterol level in plasma increases, resulting in atherosclerosis. Thyroxine also increases deposition of fats in the liver, leading to fatty liver.

Thyroxine decreases plasma cholesterol level by increasing its excretion from liver cells into bile. Cholesterol enters the intestine through bile and then it is excreted through the feces.

6. ACTION ON VITAMIN METABOLISM

Thyroxine increases the formation of many enzymes. Since vitamins form essential parts of the enzymes, it is believed that the vitamins may be utilized during the formation of the enzymes. Hence, vitamin deficiency is possible during hypersecretion of thyroxine.

7. ACTION ON BODY TEMPERATURE

Thyroid hormone increases the heat production in the body, by accelerating various cellular metabolic processes and increasing BMR. It is called thyroid hormone induced thermogenesis. During hypersecretion of thyroxine, the body temperature increases greatly, resulting in excess sweating.

8. ACTION ON GROWTH

Thyroid hormones have general and specific effects on growth. Increase in thyroxine secretion accelerates the growth of the body, especially in growing children. Lack of thyroxine arrests the growth.

Thyroxine is more important to promote growth and development of brain during fetal life and first few years of postnatal life. Deficiency of thyroid hormones during this period leads to mental retardation.

9. ACTION ON BODY WEIGHT

Thyroxine is essential for maintaining the body weight. Increase in thyroxine secretion decreases the body weight and fat storage. Decrease in thyroxine secretion increases the body weight because of fat deposition.

10. ACTION ON BLOOD

Thyroxine accelerates erythropoietic activity and increases blood volume

11. ACTION ON CVS

Increased contractility, heart rate & cardiac output

16. ACTION ON SLEEP

Normal thyroxine level is necessary to maintain normal sleep pattern.

17. ACTION ON SEXUAL FUNCTION

Normal thyroxine level is essential for normal sexual function.

In men, hypothyroidism leads to complete loss of libido (sexual drive) and hyperthyroidism leads to impotence.

In women, hypothyroidism causes menorrhagia and polymenorrhea . In some women, it causes irregular menstruation and occasionally amenorrhea.

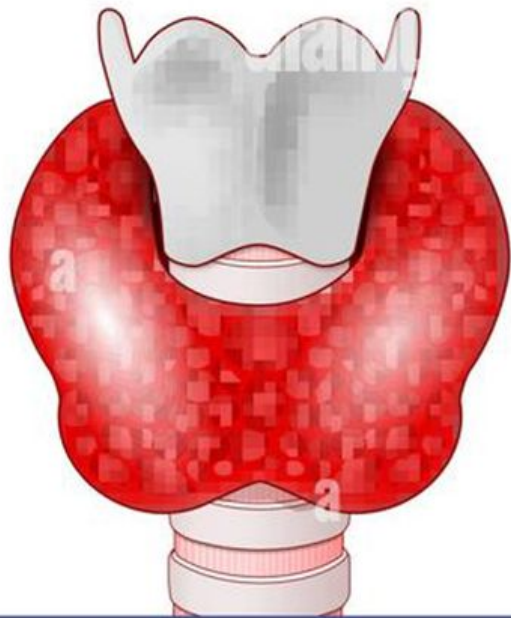
Hyperthyroidism in women leads to oligomenorrhea and sometimes amenorrhea

Actions of thyroid hormones

- **Brain** - growth & development of nervous system
- **Bone Tissue** growth – linear growth & maturation of bones
- **CVS** - increased contractility, heart rate cardiac output
- **GUT** - increased absorption of nutrients, increased motility
- **Liver** - increased gluconeogenesis glycogenolysis
- **Adipose tissue** – increased lipolysis
- **Muscle** – increased protein catabolism in skeletal muscle
- **Kidney** - increased erythropoietin synthesis
- **Respiration** - increased central stimulation of respiration
- **Energy metabolism** - increased BMR, increased oxygen consumption, increased heat production stimulation of Na-K-ATPase

APPLIED PHYSIOLOGY

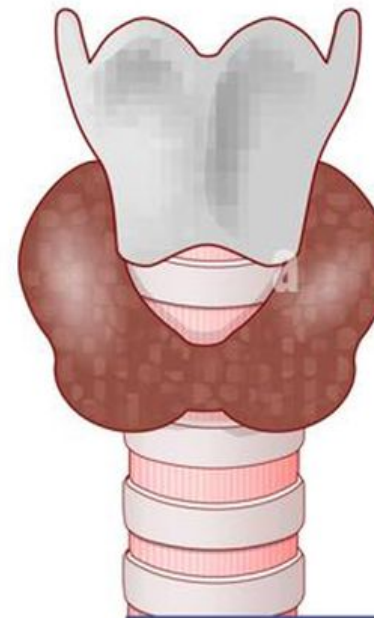
THYROID GLAND



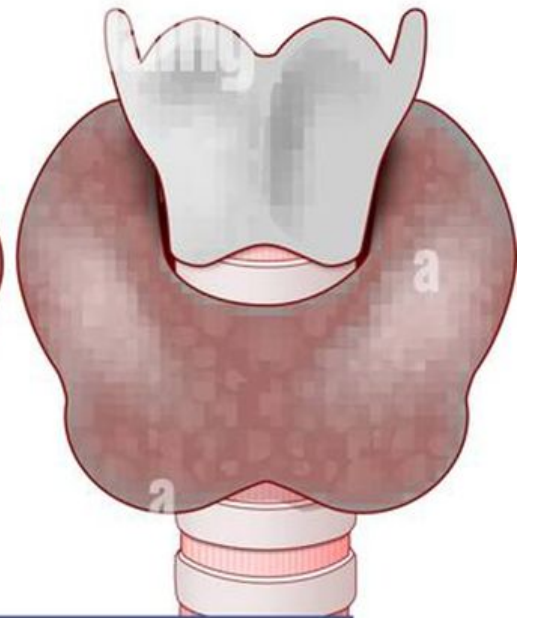
Hyperthyroidism



Normal



Hypothyroidism



DISORDERS OF THYROID GLAND

HYPERTHYROIDISM

Increased secretion of thyroid hormones is called hyperthyroidism.

Causes of Hyperthyroidism

Hyperthyroidism is caused by:

- 1. Graves' disease**
- 2. Thyroid adenoma**

Graves disease

- Most common cause of hyperthyroidism
- Also known as exophthalmic goiter
- Thyroid is diffusely enlarged
- Protrusion of the eyeballs
- Autoimmune disorder
- Circulating antibodies formed against TSH receptors which activate the receptors and make gland hyperactive.
- 4 types of antibodies-TSH stimulating, TSH-blocking, Tg antibody, Tpo antibody
- ↑ T3 ,T4 levels
- Inhibition of the TSH

Grave's disease

Presentation:

- Palpitation
- Tachycardia
- ↑ BP
- ↑ sweating
- Tremors
- ↑ appetite
- Wt loss
- Protruding eyeballs

Graves Disease.



Signs and Symptoms of Hyperthyroidism

1. Intolerance to heat as the body produces lot of heat due to increased basal metabolic rate caused by excess of thyroxine
2. Increased sweating due to vasodilatation
3. Decreased body weight due to fat mobilization
4. Diarrhea due to increased motility of GI tract
5. Muscular weakness because of excess protein catabolism
6. Nervousness, extreme fatigue, inability to sleep, mild tremor in the hands and psychoneurotic symptoms such as hyper excitability, extreme anxiety or worry.

All these symptoms are due to the excess stimulation of neurons in the central nervous system

7. Toxic goiter
8. Oligomenorrhea or amenorrhea
9. Exophthalmos
10. Polycythemia
11. Tachycardia and atrial fibrillation
12. Systolic hypertension
13. Cardiac failure

Hyperthyroidism signs and symptoms.

- Nervousness
- Wt loss
- Hyperphagia
- Heat intolerance
- Increased pulse pressure
- Fine tremors on outstretched fingers
- Warm soft skin
- Sweating
- Increased BMR

HYPOTHYROIDISM

Decreased secretion of thyroid hormones is called hypothyroidism. Hypothyroidism leads to myxedema in adults and cretinism in children.

Myxedema

Myxedema is the hypothyroidism in adults, characterized by generalized edematous appearance.

Causes for myxedema

Myxedema occurs due to diseases of thyroid gland, genetic disorder or iodine deficiency. In addition, it is also caused by deficiency of thyroid-stimulating hormone or thyrotropin-releasing hormone.

Common cause of myxedema is the autoimmune disease called Hashimoto's thyroiditis, which is common in late middle-aged women. In most of the patients, it starts with glandular inflammation called thyroiditis caused by autoimmune antibodies. Later it leads to destruction of the glands.

Signs and symptoms of myxedema

Myxedema signs and symptoms include a severe drop in body temperature, altered mental status, and extreme swelling of the skin caused by advanced, untreated hypothyroidism. This condition can lead to a life-threatening emergency known as myxedema coma, which requires immediate medical care



Difference Between Hypothyroidism and Hyperthyroidism



Fatigue



Weight Gain



Cold Sensitivity



Dry Skin & Hair Loss



Constipation



Hypothyroidism

Underactive Thyroid

VS



Nervousness & Anxiety



Weight Loss



Heat Sensitivity



Rapid Heartbeat & Palpitations



Frequent Bowel Movements



Hyperthyroidism

Overactive Thyroid

Other general features of hypothyroidism in adults are:

1. Anemia
2. Fatigue Profound weakness and constant lack of energy
3. 4. Menorrhagia and polymenorrhea
5. Decreased cardiovascular functions such as reduction in rate and force of contraction of the heart, cardiac output and blood volume
6. Increase in body weight
7. Digestive Slowdown: Severe constipation or reduced intestinal movement.
8. Mental sluggishness
9. Depressed hair growth
10. Cold intolerance

HYPOTHYroidism vs. HYPERthyroidism

LOW & SLOW

HOT & HIGH



Cretinism

- Children with hypothyroidism from birth are called cretins
- They have reduced growth and mentally retarded.
- Potbellies
- Enlarged protruding tongue

Causes-

Maternal iodine deficiency

Fetal thyroid dysgenesis

Inborn errors of thyroid synthesis

Maternal antibodies that cross placenta

Fetal hypo pituitary hypothyroidism

cretinism



GOITER

Goiter means enlargement of the thyroid gland. It occurs both in hypothyroidism and hyperthyroidism.

TREATMENT FOR THYROID DISORDERS

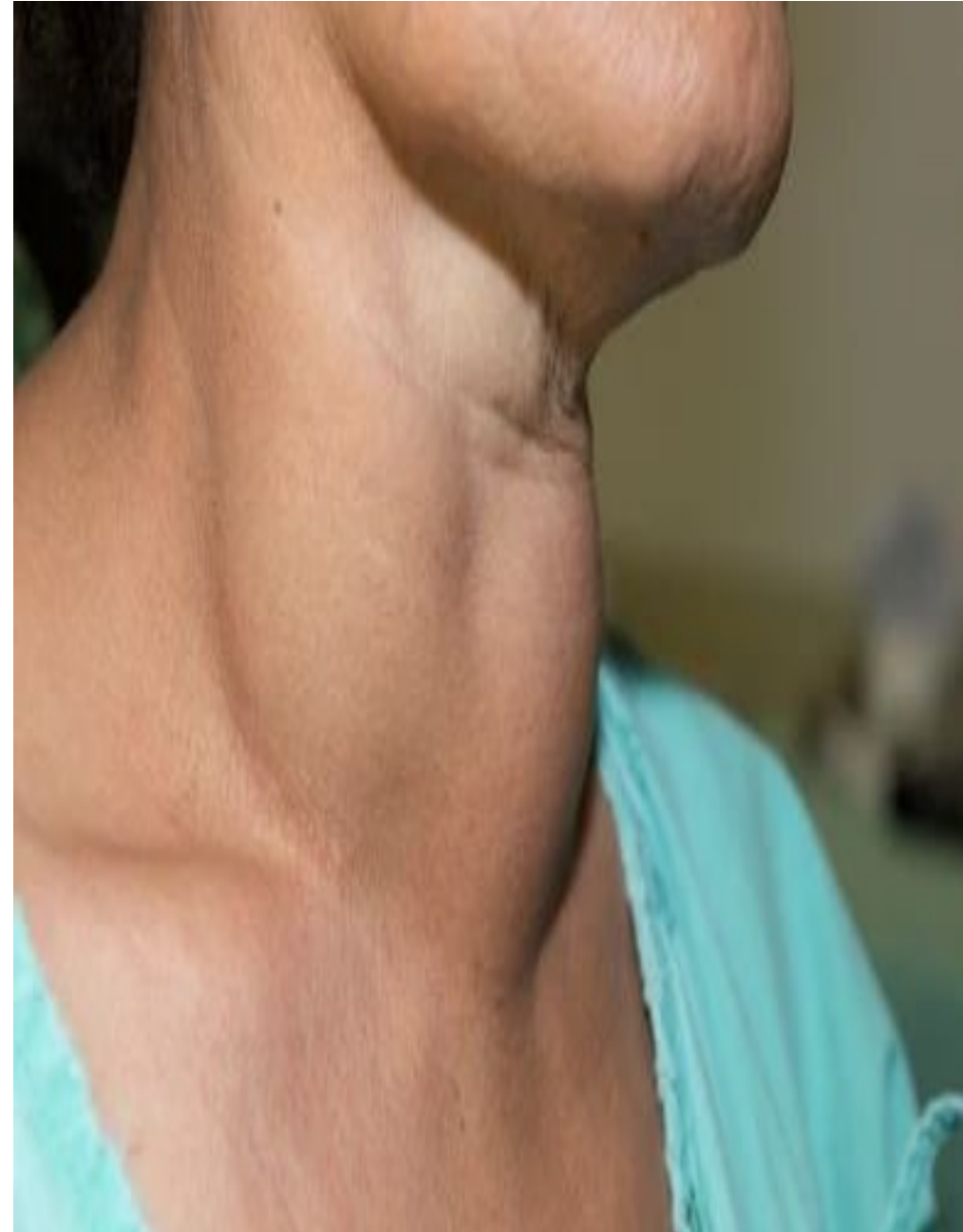
1. By using Antithyroid Substances

Antithyroid substances are the drugs which suppress the secretion of thyroid hormones. Hyperthyroidism in early stage can be treated by antithyroid substances.

Three well-known antithyroid substances are:

- i. Thiocyanate
- ii. Thioureylenes
- iii. High concentration of inorganic iodides.

2. By Surgical Removal



Goiter..



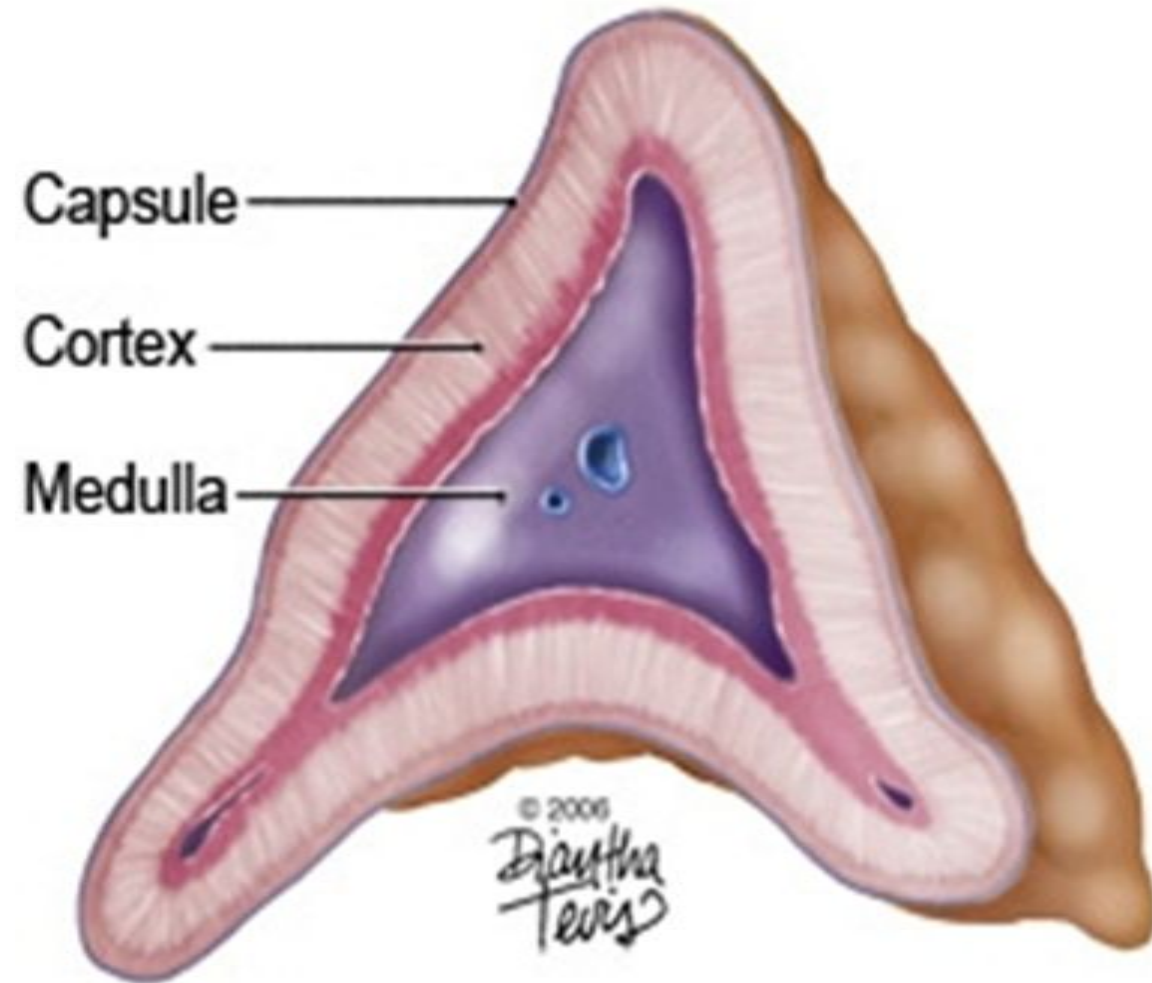
Pathological Changes in Thyroid Hormone Secretion



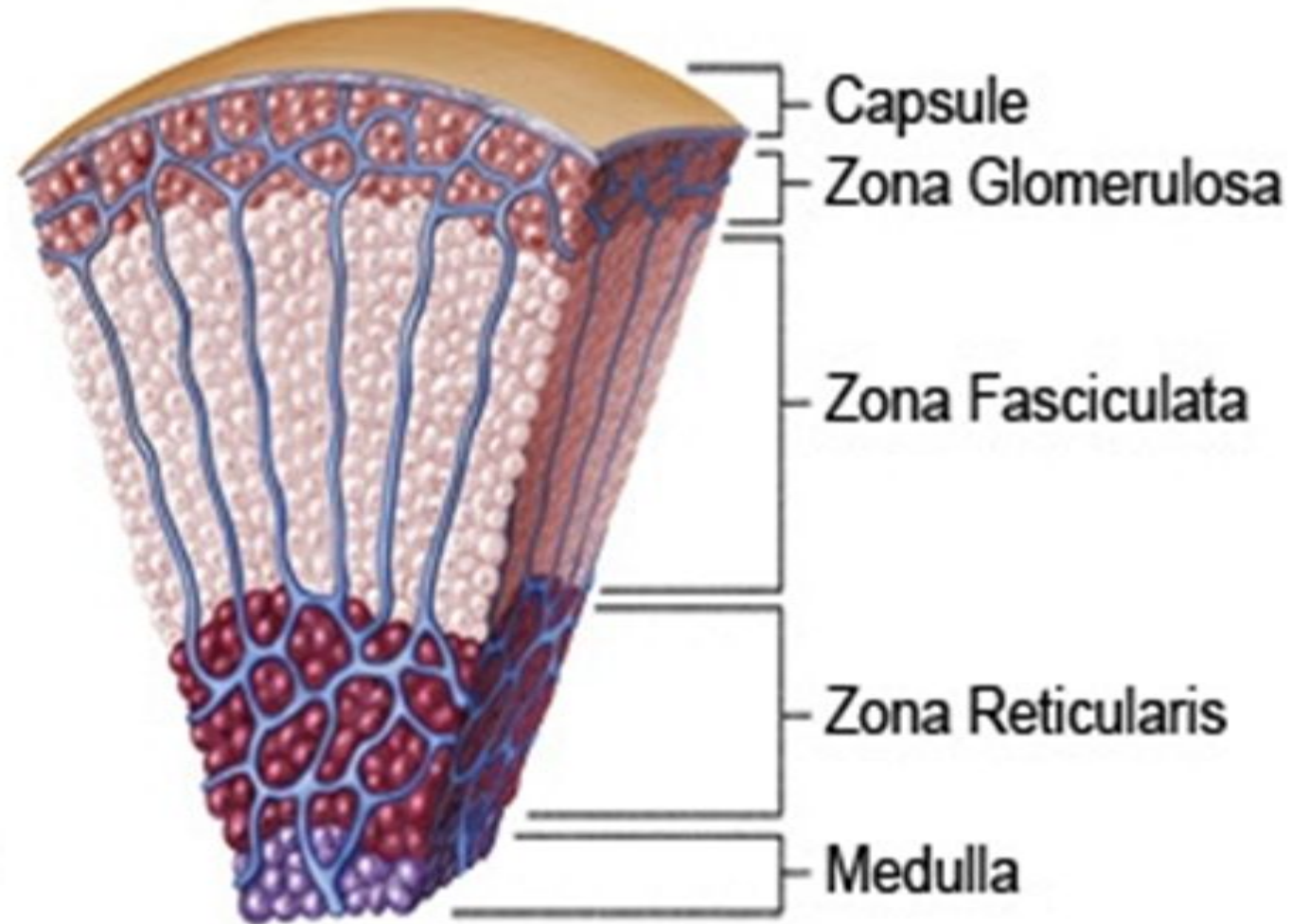
	T4	TSH	TRH
1. *Primary hypothyroidism	↓	↑	↑
2. Pituitary hypothyroidism (secondary)	↓	↓	↑
3. Hypothalamic hypothyroidism (tertiary)	↓	↓	↓
4 Pituitary hyperthyroidism (secondary)	↑	↑	↓
5 . Grave's disease (autoimmune)	↑	↓	↓

Adrenal gland: Structure, Hormones and clinical disorders

Transverse Section

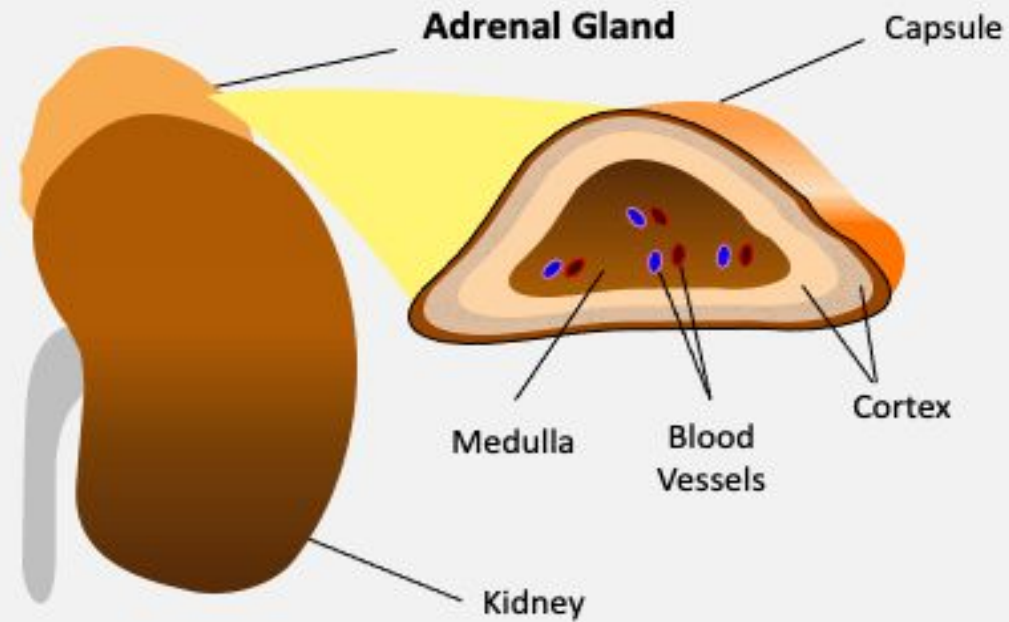


Microscopic Section



ADRENAL GLAND

What is Adrenal Gland?



Adrenal Glands are small glands attached to the top of each kidney. The human body has two adrenal glands that release chemical messengers called hormones into the bloodstream. These hormones travel via the blood to have an action in other parts of the human body

Adrenal glands are called the 'life-saving glands' or 'essential endocrine glands'.

It is because the absence of adrenocortical hormones causes death within 3 to 15 days and absence of adrenomedullary hormones, drastically decreases the resistance to mental and physical stress.

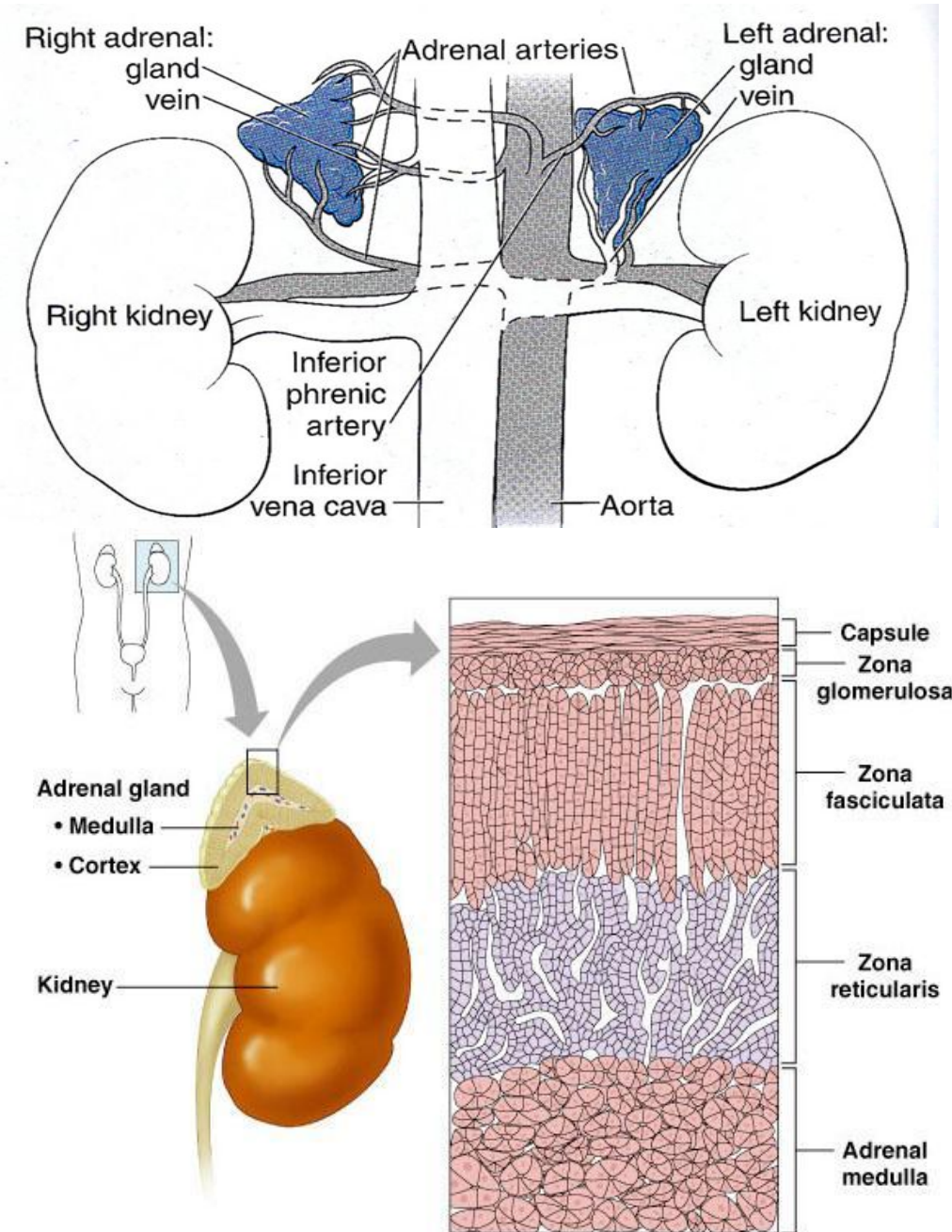
They are small, triangular organs sitting above your kidneys that produce vital hormones to control metabolism, blood pressure, and your response to stress.

Anatomy of adrenal glands

Localization: At the top of the kidney and weighing approximately 5 g each.

Each adrenal gland is composed of two endocrine components:

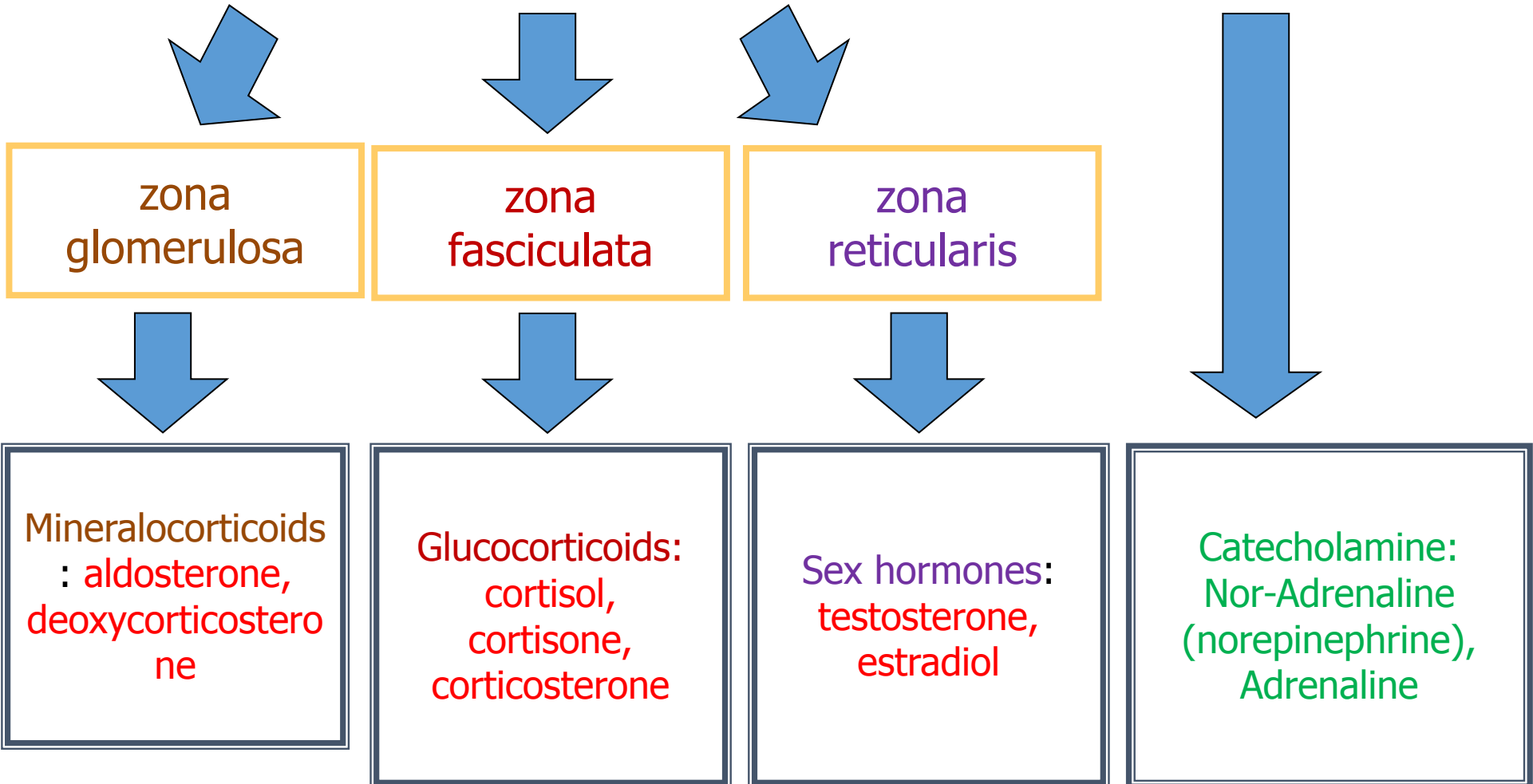
- **Cortex** (outer part) that constitutes the 80% of the gland.
 - The cortex consists of three zones.
 - The medulla and each of the zones in the cortex each produce different hormones that serve a variety of functions in your body.
- **Medulla** (inner part) that constitutes 20% of the gland



Adrenal glands

Cortical substance

Medullary substance



Functions of the Adrenal Gland

Maintaining homeostasis Role in the regulation:

- of the body's adaptive response to stress
- in the maintenance of body water, Na and K balance
- in the control of blood pressure

The main hormones

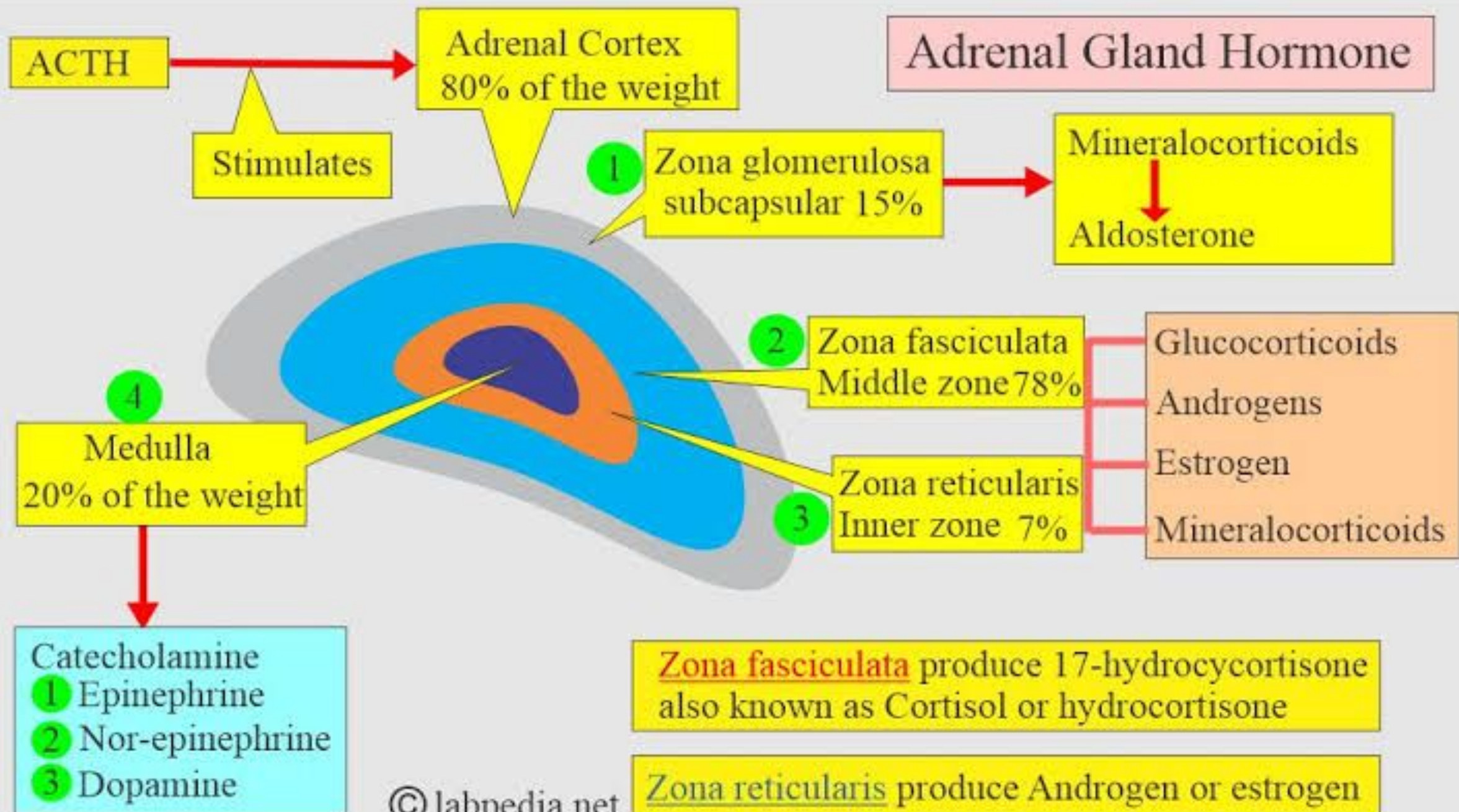
□ Adrenal Cortex

The Steroid hormones: mineralocorticoids, glucocorticoids, and androgens

➤ Adrenal Medulla

The Catecholamine (Norepinephrine, Epinephrine)

The AG has 2 different embryologic origins, which influence the mechanisms that control hormone production by each of the 2 components.



Part -1

Adrenal Cortex

Functional Anatomy and Zonation

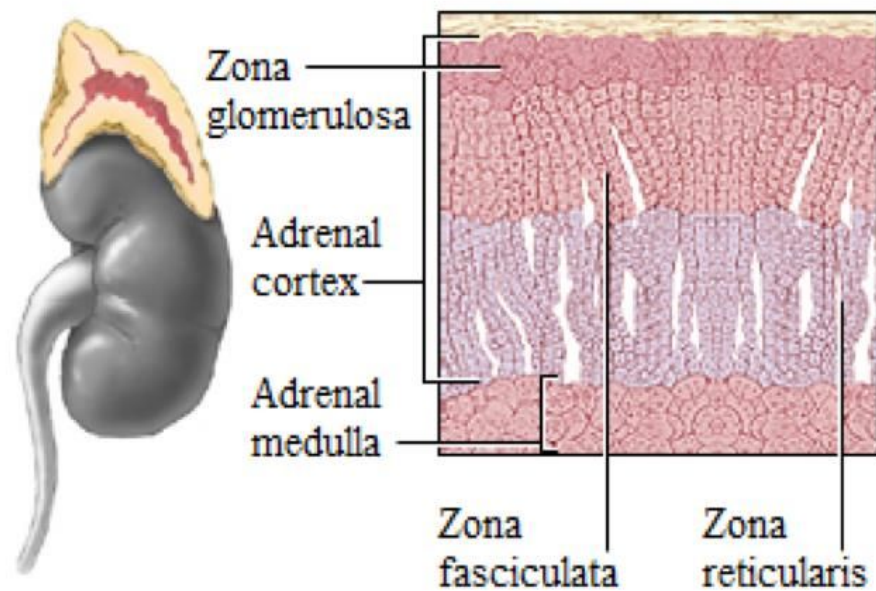
The adrenal cortex contains three histologically distinct zones (from outside to inside):

- zona glomerulosa
- zona fasciculata
- zona reticularis

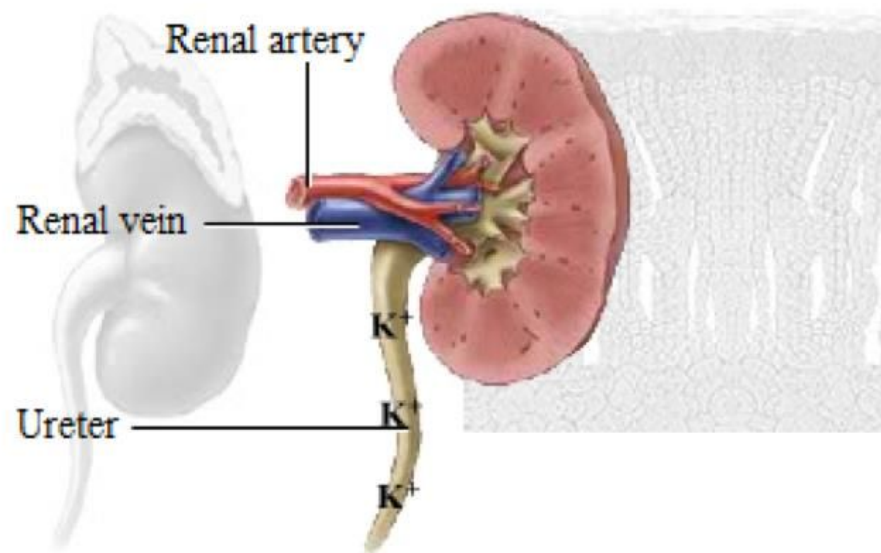
Hormones secreted by the adrenal cortex include:

- Mineralocorticoids (Aldosterone)
- Glucocorticoids
- Adrenal androgens OR sex Hormones

The Adrenal Cortex

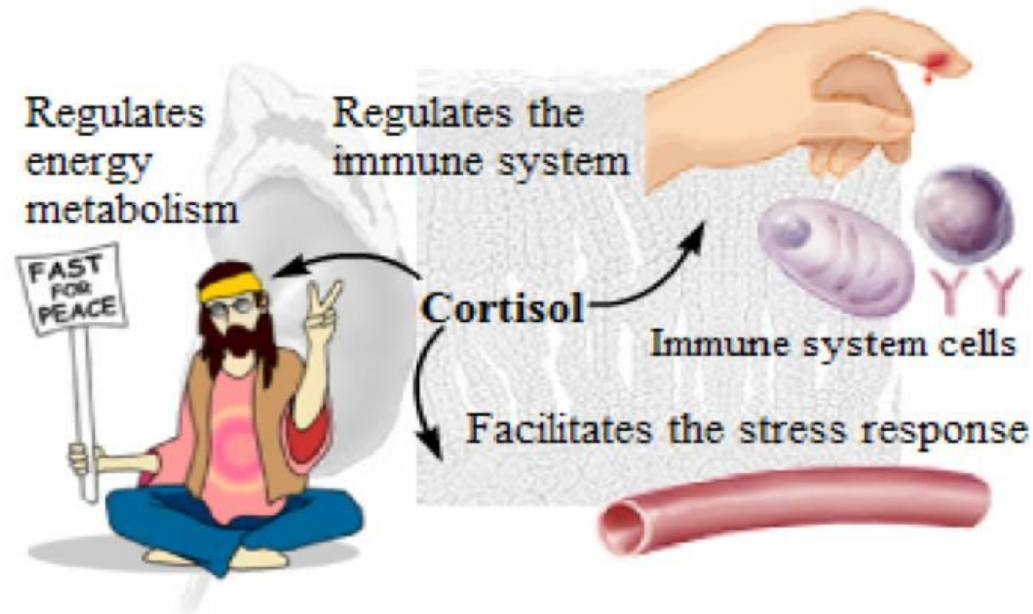


The cortex secretes three families of steroid hormones: **mineralocorticoids**, glucocorticoids, and **androgens**. Note that each is produced by a different group of cells. The medulla secretes amine hormones known as **catecholamines**.



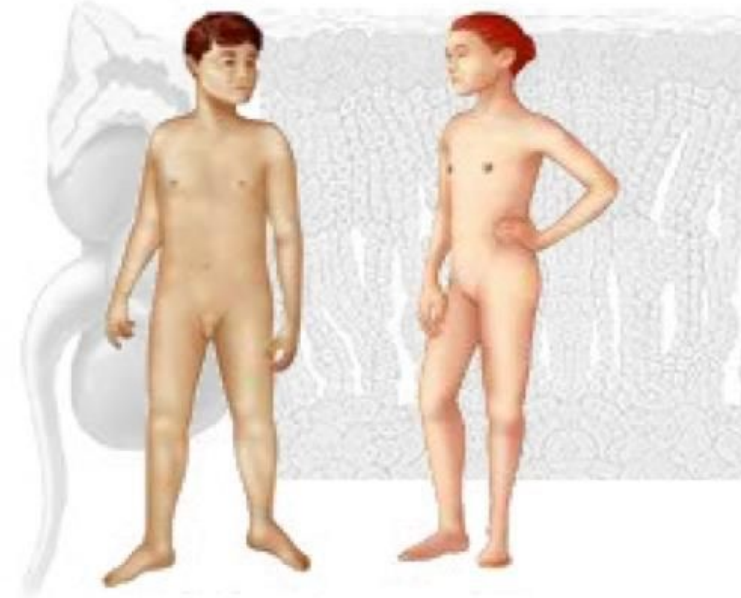
Cells of the zona glomerulosa produce hormones called mineralocorticoids because they affect salt balance. **Aldosterone** is the most important. Among its functions are sodium retention and potassium excretion by the kidneys.

The Adrenal Cortex (cont.)



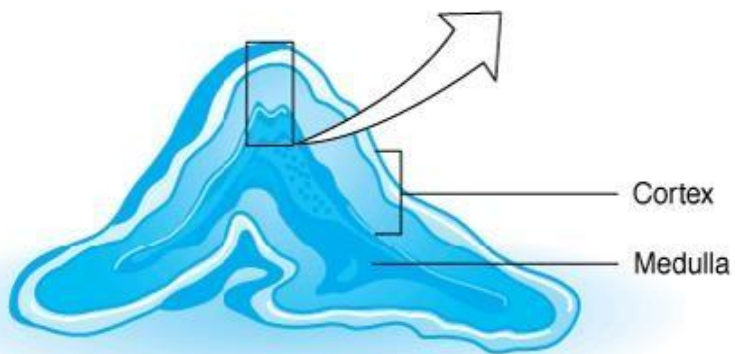
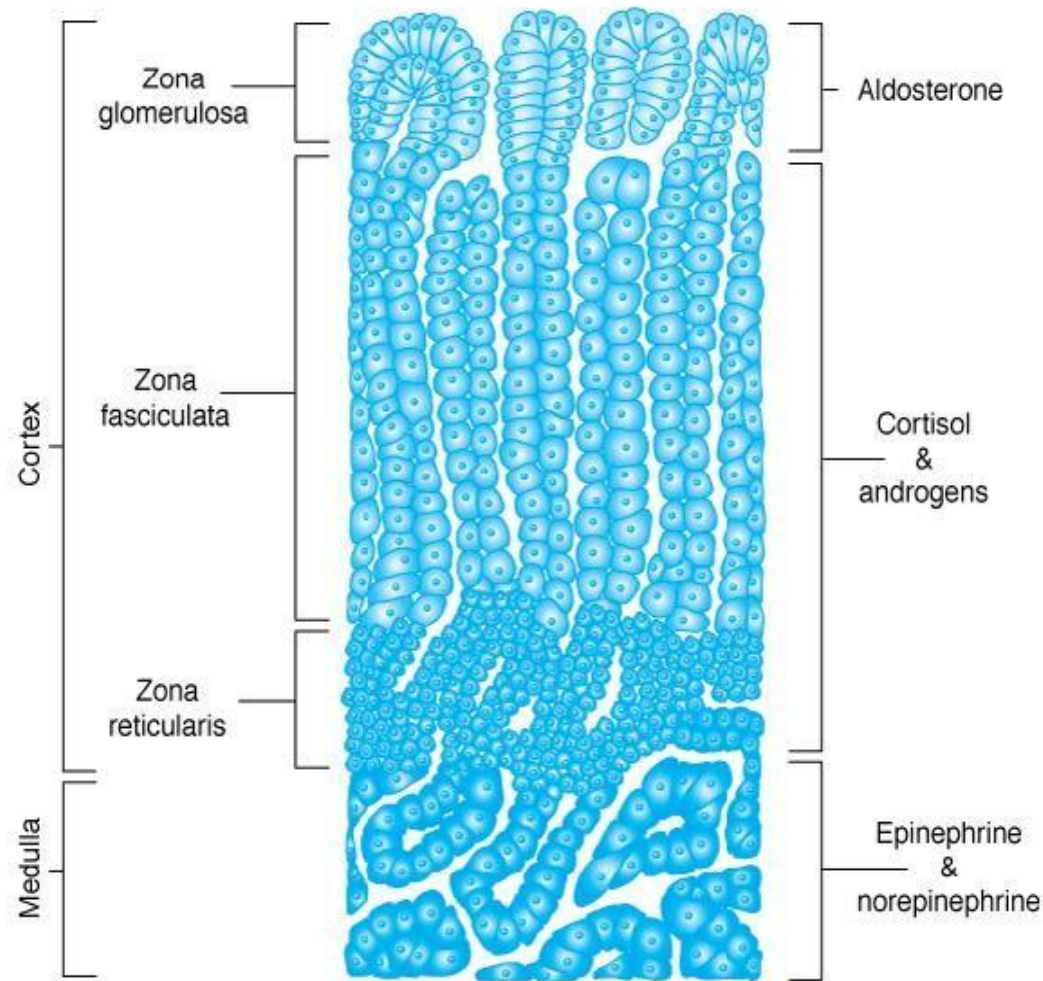
Cells of both zona fasciculata and zona reticularis produce a family of hormones called glucocorticoids because they regulate glucose metabolism. This family includes **cortisol**, **cortisone**, and **corticosterone**.

Only cortisol is secreted in significant amounts in humans. In addition to regulating energy metabolism, cortisol regulates the immune system and facilitates the stress response.



Cells of both zona fasciculata and zona reticularis also produce a family of hormones called androgens. Androgens are male sex hormones that include **androstenedione** and testosterone.

Active secretion at puberty in both sexes causes the early growth spurt and appearance of axillary and pubic hair. Androgens are also responsible for sex drive in females.



Zonation

The **mineralocorticoid** synthesized in the zona glomerulosa of the adrenal cortex.

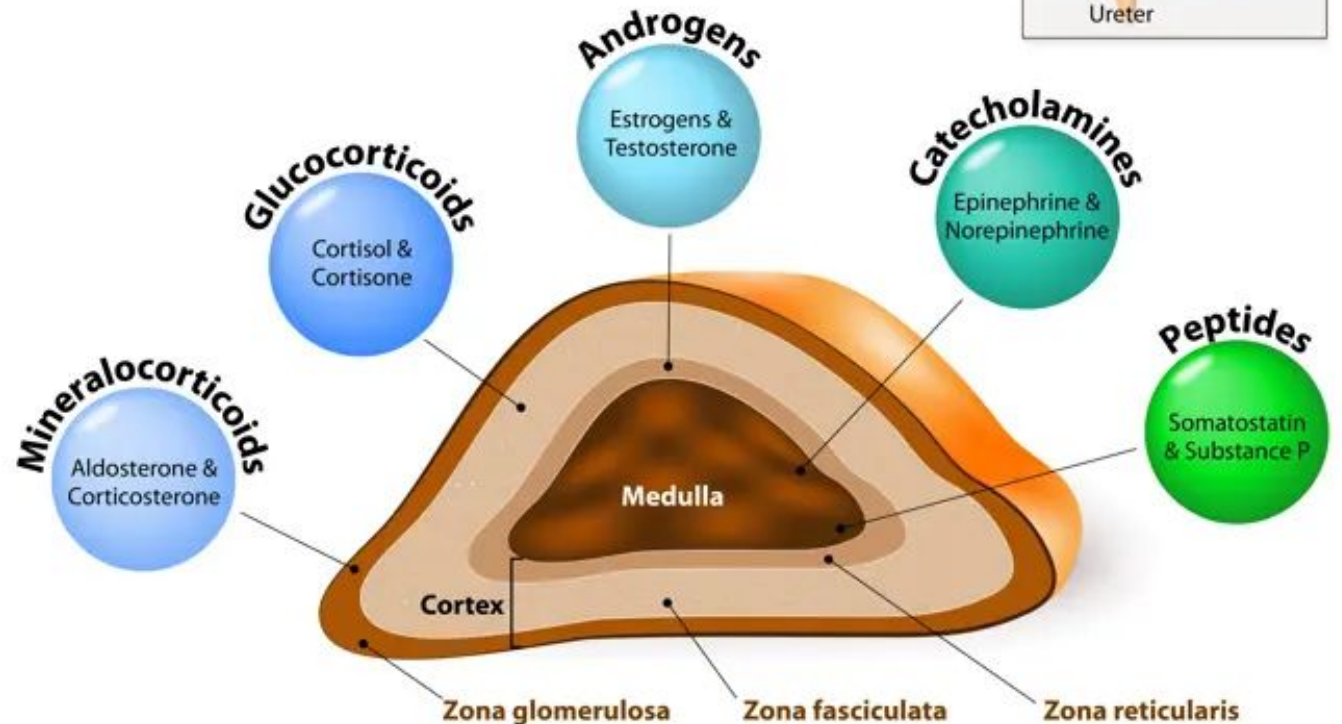
The **glucocorticoids** cortisol and corticosterone, zona fasciculata

The **androgen** dehydroepiandrosterone, or **sex hormones** are synthesized in the zona reticularis of the adrenal cortex.

Hormones of the The Adrenal Cortex

- All adrenal cortex hormones are called **corticosteroid**, because all synthesize from steroid cholesterol (LDL-cholesterol)

ADRENAL GLAND (hormones)



Hormones of the The Adrenal Cortex

Hormones produced by the adrenal cortex are referred to as Corticosteroids
Different corticosteroids are produced in each of the **three layers**:

Zona glomerulosa –

Mineralocorticoids (Mainly Aldosterone)

Zona fasciculata –

glucocorticoids (mainly cortisol and corticosterone)

Zona reticularis –

Gonadocorticoids (Androgens Sex Hormones)

Mineralocorticoid

Mineralocorticoids are the corticosteroids that act on the minerals (electrolytes), particularly sodium and potassium.

Mineralocorticoid Maintains Na^+ balance, by reducing excretion of sodium from the body

It stimulates reabsorption of Na^+ by the kidneys and K^+ excretion.

Main Hormones are:

- ❑ Aldosterone-90 %
- ❑ Deoxy-corticosterone
- ❑ Corticosterone
- ❑ 9 alpha fluorocortisol (synthetic Aldosterone) • Cortisol
- ❑ Cortisone

Functions of the Aldosterone

The functions of aldosterone, a hormone primarily produced by the adrenal glands, are crucial for maintaining the body's electrolyte balance, fluid volume, and blood pressure. Here are some key functions of aldosterone:

On Kidneys Sodium Reabsorption

Aldosterone, a mineralocorticoid, causes the Kidneys to conserve Sodium ions and thus water, and to excrete Potassium ions

Aldosterone Increases Renal Tubular Reabsorption of Sodium and Secretion of Potassium by acting on the distal tubules and collecting ducts of the nephrons in the kidneys, increasing the activity of sodium channels in these tubules.

High levels of aldosterone decreases Na loss and increase K loss. By retaining sodium, aldosterone helps to regulate extracellular fluid volume and blood pressure.

Potassium Excretion

In addition to promoting sodium reabsorption, aldosterone stimulates the excretion of potassium ions in the kidneys. It does so by increasing the secretion of potassium into the renal tubules, which is then eliminated from the body through urine. This process helps to maintain appropriate levels of potassium in the bloodstream, preventing hyperkalemia (high potassium levels) which can have serious health consequences.

Hydrogen Ion Excretion

Aldosterone also facilitates the excretion of hydrogen ions (H^+) in the kidneys. By promoting the secretion of hydrogen ions into the renal tubules, aldosterone indirectly helps to regulate blood pH and maintain acid-base balance within the body.

Functions of the Aldosterone

Water Retention

The reabsorption of sodium ions by aldosterone in the kidneys leads to increased osmolarity of the extracellular fluid. This osmotic effect causes water to be retained along with sodium, thereby expanding the volume of extracellular fluid.

Regulation of Blood Pressure

Through its actions on sodium and water retention, aldosterone plays a crucial role in regulating blood pressure.

Aldosterone is secreted in response to decreasing blood volume and blood pressure as a result of changes in the kidney.

Person develops hypertension which persist as long as exposed to high levels of Aldosterone.

Response to Stress and Dehydration

Aldosterone secretion is stimulated by various factors, including decreased blood volume, low blood pressure, and increased potassium levels.

In response to stress, dehydration, or other conditions that threaten fluid and electrolyte balance, aldosterone secretion increases to conserve sodium and water and restore normal blood volume and pressure.

(On sweat and salivary glands)

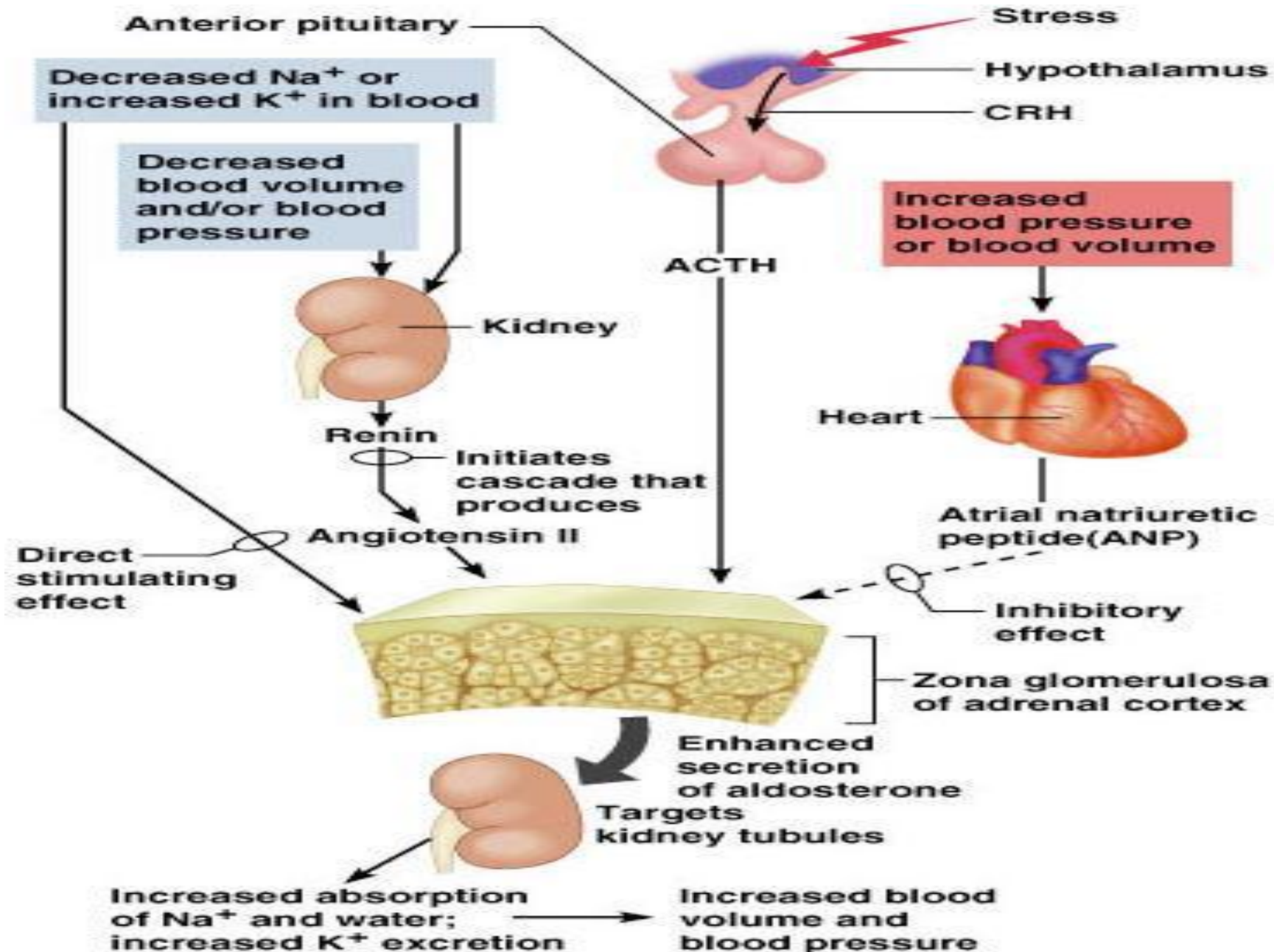
Aldosterone stimulates Sodium and Potassium transport in Sweat glands, Salivary glands and Intestinal Epithelial cells. Aldosterone increases the reabsorption of NaCl and secretion of K by ducts to conserve body salts in hot environment.

Regulation Of Aldosterone

Aldosterone secretion is stimulated by four factors:

- Rising blood levels of K^+
- Decreasing blood volume or pressure (renin-angiotensin system).
- Low blood Na^+
- ACTH

The Four Mechanisms of Aldosterone Secretion



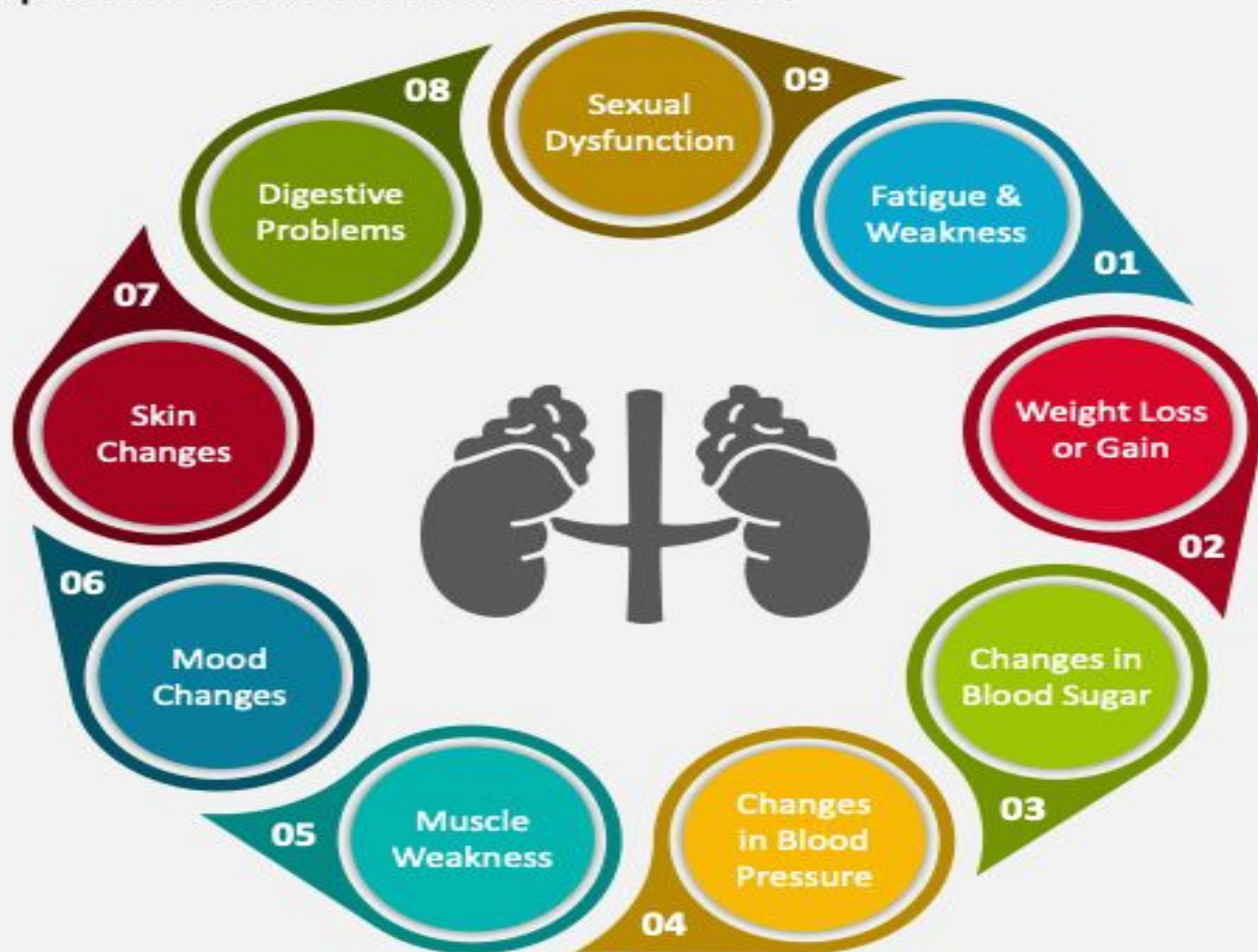
Regulation of Aldosterone Secretion

The Four Mechanisms of Aldosterone Secretion

- Increased potassium ion concentration in the extracellular fluid greatly *increases* aldosterone secretion.
- Increased angiotensin II concentration in the extracellular fluid also greatly *increases* aldosterone secretion.
- Increased sodium ion concentration in the extracellular fluid *very slightly decreases* aldosterone secretion.
- ACTH from the anterior pituitary gland is necessary for aldosterone secretion but has little effect in controlling the rate of secretion in most physiological conditions.

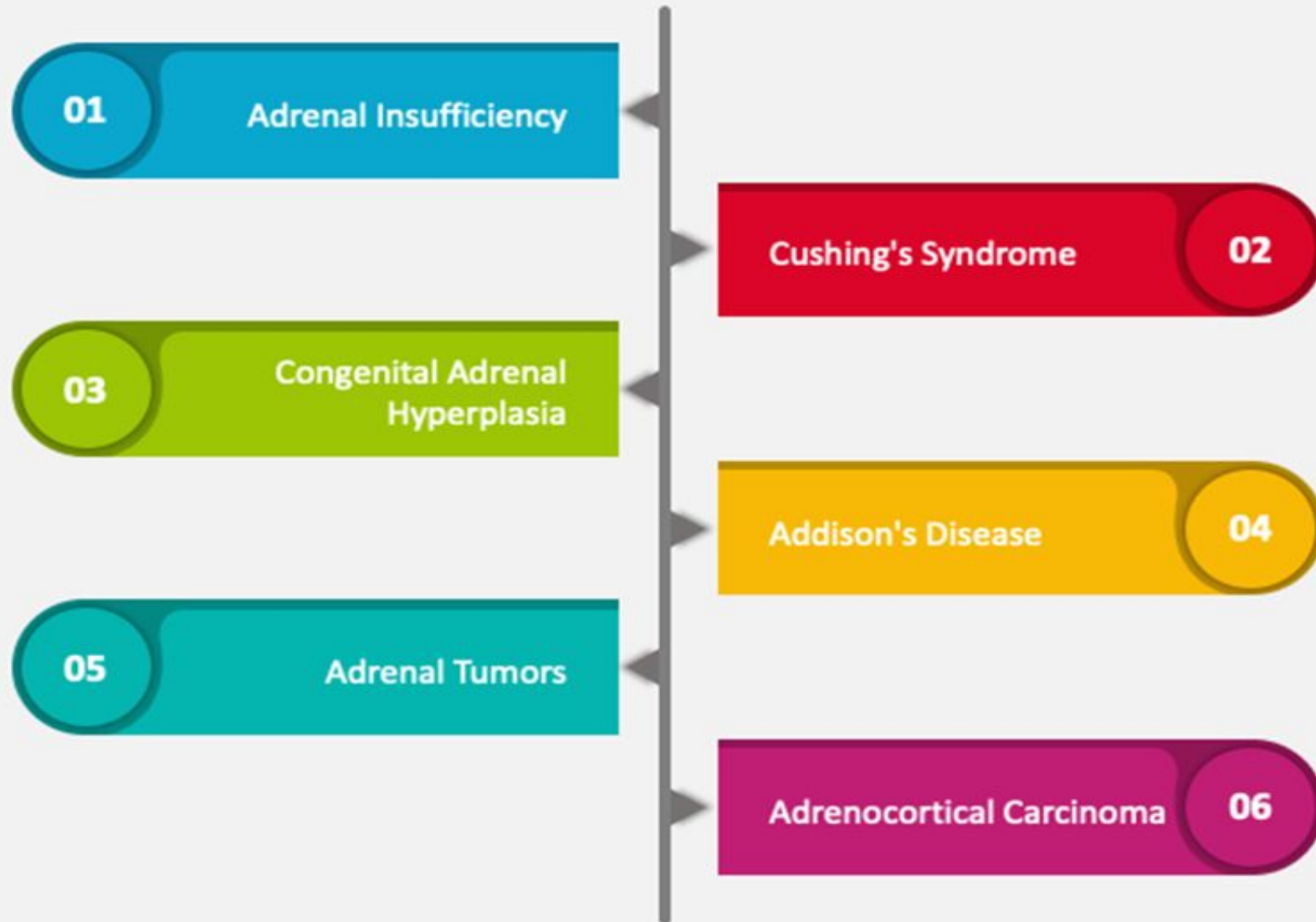
ADRENAL GLAND

Signs and Symptoms of Adrenal Gland Problems



ADRENAL GLAND

Disorders Affecting the Adrenal Glands



ADRENAL GLAND

Prevention of the Adrenal Gland



Part -2

Adrenal Medulla

ADRENAL MEDULLA

INTRODUCTION

Medulla is the inner part of adrenal gland and it forms 20% of the mass of adrenal gland.

It is made up of interlacing cords of cells known as chromaffin cells.

Chromaffin cells are also called pheochrome cells or chromophil cells.

These cells contain fine granules which are stained brown by potassium dichromate.

The adrenal medulla secretes stress-response hormones into the bloodstream

Location and Structure

Position: It sits at the center of each adrenal gland, located just above the kidneys.

Adrenal Medullary Hormones

Adrenal Medullary hormones are involved in control of

- ☐ Heart rate,
- ☐ Blood pressure,
- ☐ Blood vessel constriction in various organs like skin, gastrointestinal tract
- ☐ Dilation of blood vessel in skeletal muscles.

Adrenal Medulla

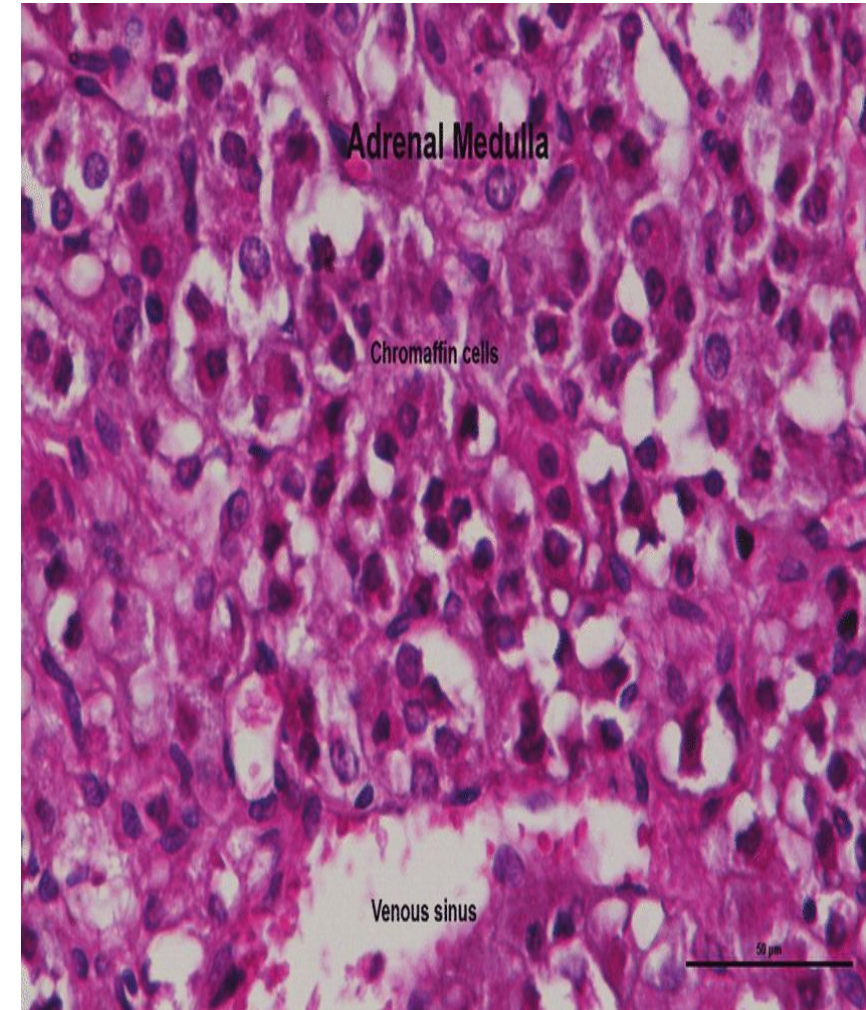
HORMONES OF ADRENAL MEDULLA

Adrenal medullary hormones are the amines derived from catechol and so these hormones are called catecholamines.

Catecholamines secreted by adrenal medulla

1. Adrenaline or epinephrine - 80%
2. Noradrenaline or norepinephrine - 20%
3. Dopamine

Secretion of these hormones causes Blood to be diverted to the Brain ,Heart and skeletal muscles.



Functions of catecholamines

- Glycogenolysis in Liver and skeletal muscles (can lead to hyperglycemia which increases blood glucose level).
- Increase heart rate and blood pressure.
- Causes vasoconstriction of blood vessels
- Metabolization of free fatty acids
- Increase Metabolic rate
- Increase O₂ consumption

Adrenal Medulla

The function of adrenal medulla is governed by nervous system and the chromaffin cells release the hormones responsible for various metabolic reactions.

Release of Adrenaline and Noradrenaline is triggered by nervous stimulation in response to physical or mental stress.

Adrenaline (epinephrine) acts primarily as a stress hormone released by the adrenal glands, whereas noradrenaline (norepinephrine) functions mainly as a neurotransmitter in the nervous system while also serving as a hormone

The three primary catecholamines in the body are:

- **Dopamine:**

Controls movement, memory, mood, and the brain's reward system.

- **Norepinephrine (Noradrenaline):**

Increases heart rate and blood pressure to help the body respond to stress.

- **Epinephrine (Adrenaline):**

Powers the immediate fight-or-flight response by sending extra blood to the heart, lungs, and muscles

Dopamine

- Dopamine is a type of neurotransmitter and hormone. It is secreted from the adrenal medulla as well as from the hypothalamus.
- It plays a role in many important body functions, including, memory and pleasurable reward and motivation mood, learning, sleep, memory, lactation, and involuntary movement.
- High or low levels of dopamine are associated with several mental health and neurological illness.

Epinephrine

Epinephrine (Adrenaline) Main Role:

Acts primarily as a hormone released by the adrenal glands.

Heart and lungs: Increases heart rate and opens airways to improve breathing.

Metabolism: Breaks down glycogen into glucose to spike blood sugar and energy.

Medical use: Treats severe allergic reactions (anaphylaxis), severe asthma, and cardiac arrest.

Abnormal levels are linked to sleep disorders, anxiety, hypertension and lowered immunity.

Epinephrine's major action is in its role as a hormone. Epinephrine is released by your adrenal glands in response to stress.

Physical Indications of Fight or Flight Response



dilated pupils

trembling



pale or flushed skin

rapid heart beat
and breathing

Noradrenaline

Norepinephrine (Noradrenaline) Main Role:

Acts primarily as a neurotransmitter in the sympathetic nervous system and partly as a hormone.

Blood pressure: Constricts blood vessels to maintain stable blood pressure during stress.

Brain function: Increases alertness, arousal, attention, and focus.

Medical use: Administered as an emergency injection to treat dangerously low blood pressure and shock

This will boost the supply of oxygen and glucose to the brain and muscles, and increase skeletal muscle readiness.

This prepares your body to react to stressors.

Abnormalities OR Disorders

Abnormalities of Adrenocortical Secretion

There are many types of adrenal gland disorders, including:

Addison's Disease:

a condition in which the adrenal glands don't make enough cortisol. It is due to the underproduction of cortisol and aldosterone, often due to an autoimmune response

Cushing's Syndrome:

a condition caused by too much cortisol in the body, often from taking steroid medicines for a long time. It is due to the overproduction of cortisol by the adrenal glands.

Pheochromocytoma:

a rare tumor of the adrenal medulla that produces too much adrenaline and noradrenaline.

Adenoma:

a benign tumor (not cancer) that makes too much aldosterone and may cause serious high blood pressure

Congenital Adrenal Hyperplasia (CAH):

a group of inherited disorders in which the adrenal glands don't make enough cortisol

Hypoadrenalism (Adrenal Insufficiency)

- Addison's disease results from an inability of the adrenal cortices to produce sufficient adrenocortical hormones, and this in turn is most frequently caused by primary atrophy or injury of the adrenal cortices.
- Addison's is most commonly caused by autoimmune destruction of the adrenal cortex
- Adrenal gland hypofunction is also frequently caused by tuberculosis destruction of the adrenal glands or invasion of the adrenal cortices by cancer.
- In some cases, adrenal insufficiency is secondary to impaired function of the pituitary gland, which fails to produce sufficient ACTH. W

Hypoadrenalism (Adrenal Insufficiency)

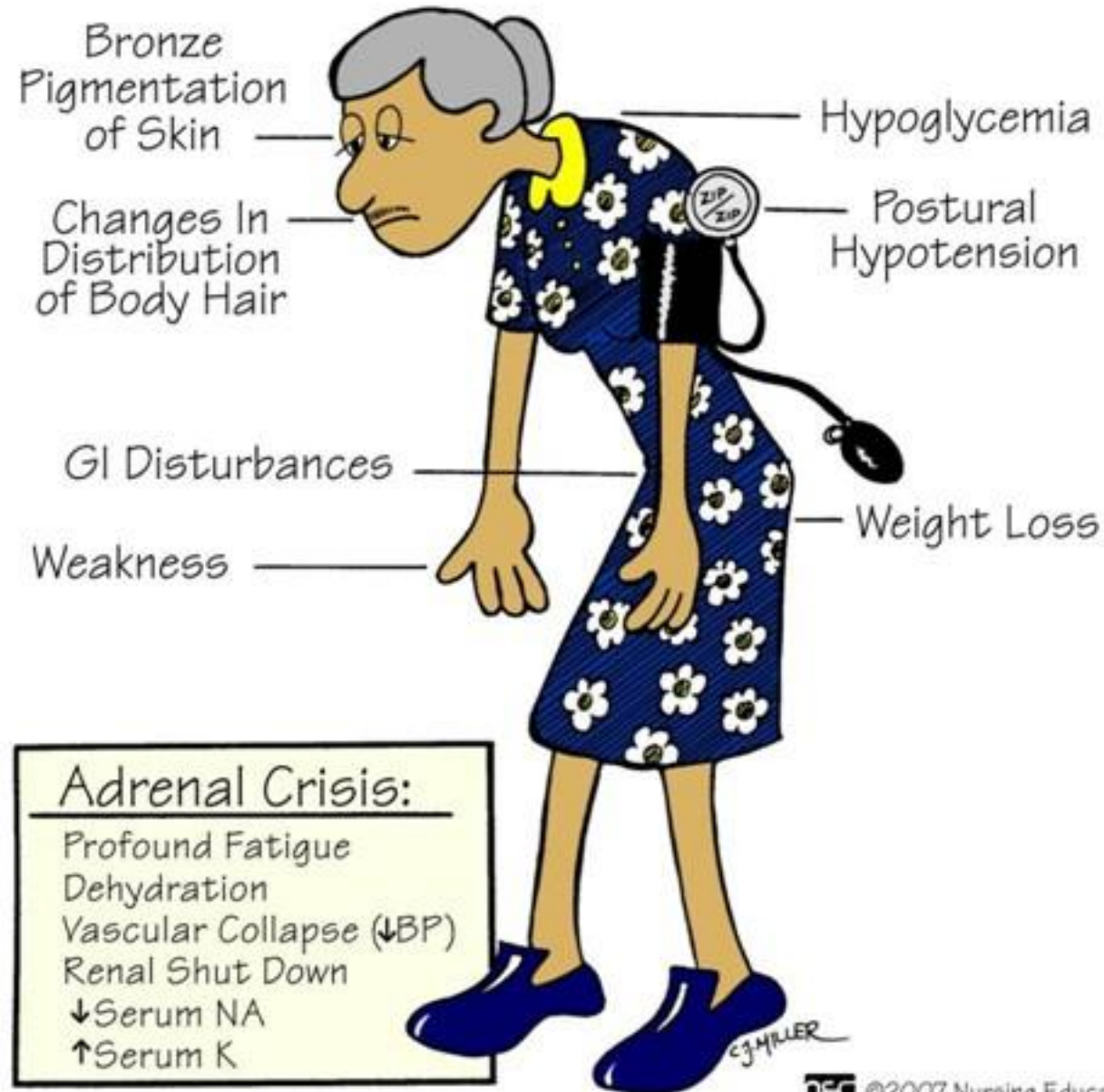
Sign and Symptoms

- Melanin Pigmentation
- Hypoglycemia
- Wt. loss, weakness, nausea and vomiting
- Decreased Axillary and Pubic hairs (caused by deficiency of the Androgen hormones)
- ECF volume contraction □ hypotension, hyperkalemia, and metabolic disorders.



due
in

ADDISON'S DISEASE



Cushing's Syndrome

Causes:

- Increased Plasma Levels of Glucocorticoids (Cortisol) Produce Cushing's Syndrome and is characterized by high plasma levels of ACTH and cortisol
- Cushing's syndrome can also occur when large amounts of glucocorticoids are administered over prolonged periods for therapeutic purposes. For example, patients with chronic inflammation associated with diseases such as rheumatoid arthritis are often treated with glucocorticoids and may develop some of the clinical symptoms of Cushing's syndrome.

- Mobilization of fat from the lower part of the body, with concomitant extra deposition of fat in the thoracic and upper abdominal regions, giving rise to a buffalo torso or buffalo hump.
- An edematous appearance of the face, acne and hirsutism (“moon face”)
- Hypertension, hyperglycemia, severe muscle weakness, large purplish striae, severe osteoporosis with consequent weakness of the bones



Cushing's Syndrome



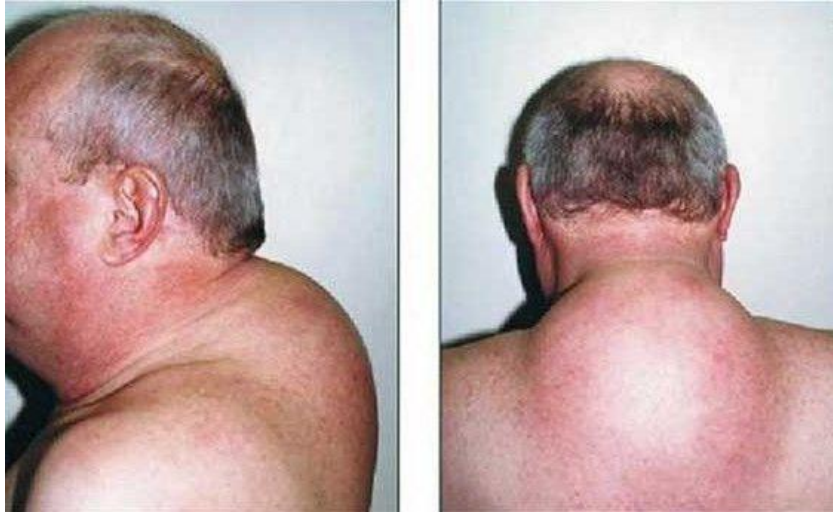
(a) Patient before onset.

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(b) Same patient with Cushing's syndrome.
The white arrow shows the characteristic
"buffalo hump" of fat on the upper back.

Cushing's syndrome, Cushing's disease



Mineralocorticoid Deficiency

It results in:

- ❑ Excessive loss of sodium
- ❑ Hypovolemia
- ❑ Hypotension
- ❑ Excessive potassium retention and hyperkalemia
- ❑ Mild acidosis

Glucocorticoid Deficiency

- Abnormal carbohydrate, fat, and protein metabolism resulting in muscle weakness, fasting hypoglycemia, and impaired utilization of fats for energy.
- Loss of appetite and weight loss.
- Poor tolerance to stress.

CONCLUSION:

Thyroid Gland

- **Location and Shape:** The Thyroid gland sits at the front of the neck under the skin. It has a small, butterfly shape.
- **Hormones Produced:** It releases T3 (triiodothyronine) and T4 (thyroxine) to control how the body uses energy. It also releases calcitonin to manage calcium levels.
- **Control Center:** The hypothalamus releases TRH, which tells the pituitary gland to release TSH (thyroid-stimulating hormone). TSH then tells the thyroid to make T3 and T4.

Adrenal Glands

- **Location and Structure:** The Adrenal gland sits on top of each kidney. Each gland has an outer cortex and an inner medulla.
- **Hormones Produced:** The cortex makes cortisol (for stress and metabolism) and aldosterone (for salt and water balance). The medulla makes adrenaline and noradrenaline for quick fight-or-flight reactions.
- **Control Center:** The hypothalamus releases CRH, which tells the pituitary gland to release ACTH. ACTH then signals the adrenal cortex to release cortisol.

Shared and Overlapping Symptoms

- **Persistent Fatigue:** Exhaustion that does not improve even after getting enough sleep or rest.
- **Weight Fluctuations:** Unexplained weight gain or trouble losing weight, as well as unexpected weight loss.
- **Mood Changes:** Increased anxiety, irritability, brain fog, or feelings of depression.
- **Temperature Sensitivity:** Frequently feeling cold or struggling to maintain a comfortable body temperature.

Distinct Adrenal Symptoms (Cortisol/Insufficiency)

- **Dizziness:** Lightheadedness or a drop in blood pressure when standing up quickly.
- **Cravings:** Strong, frequent desires for salty or sugary foods.
- **Sleep Disruption:** Feeling "wired but tired" at night, making it difficult to fall or stay asleep.
- **Skin Changes:** Dark patches or hyperpigmentation on the skin in primary adrenal insufficiency.

Distinct Thyroid Symptoms (Hypo/Hyperthyroidism)

- **Hair and Skin:** Dry skin, hair loss, or brittle, thinning hair.
- **Digestive Shifts:** Constipation with an underactive thyroid, or more frequent bowel movements with an overactive thyroid.
- **Heart Rate:** A fast or irregular heartbeat in hyperthyroidism, or a slowed metabolism overall in hypothyroidism.

Common Signs and Symptoms

- Persistent fatigue:** Ongoing low energy that does not improve with rest.
- Unexplained weight gain:** Putting on weight despite no changes in diet or exercise (common in hypothyroidism).
- Difficulty losing weight:** Struggling to shed pounds even with effort.
- Unintentional weight loss:** Losing weight rapidly without trying (common in hyperthyroidism).
- Sensitivity to cold:** Feeling chilly when others feel comfortable.
- Heat intolerance:** Sweating excessively or feeling uncomfortably hot.
- Dry skin and brittle nails:** Skin that feels rough or scaly alongside weak nails.
- Hair thinning or loss:** Noticeable hair shedding on the scalp, eyebrows, or body.
- Constipation:** Slower bowel movements and digestive sluggishness.
- Diarrhea or frequent bowel movements:** Increased digestive speed or looser stools.
- Low mood or depression:** Feelings of sadness, apathy, or persistent low spirits.
- Anxiety and nervousness:** Feeling jittery, anxious, or high-strung.
- Memory and concentration difficulties:** Experiencing "brain fog" or forgetfulness.
- Muscle weakness:** Feeling a lack of physical strength in the arms or legs.
- Joint pain and aching:** Stiffness or discomfort in the joints.
- Menstrual changes:** Heavier, lighter, or irregular periods.
- Reduced libido:** A lower sex drive than usual.
- Swelling in the neck area:** A visible or palpable enlargement known as a goiter.
- Hoarseness or voice changes:** A raspy or deeper voice due to pressure on the larynx.

Signs of Low Hormones (Adrenal Insufficiency)

When your body lacks enough adrenal hormones, like in Addison's disease, you may notice:

- **Extreme fatigue:** Tiredness that does not get better with sleep.
- **Muscle weakness:** Feeling weak in your arms and legs.
- **Weight loss:**
Losing weight without trying and having a low appetite.
- **Low blood pressure:** Feeling dizzy or lightheaded when you stand up.
- **Salt cravings:** A strong urge to eat salty foods.
- **Skin darkening:** Patches of skin getting darker, especially in skin folds or scars

Signs of High Hormones (Cushing's Syndrome)

When your body makes too much of the stress hormone cortisol, you may notice:

- **Weight gain:** Gaining weight mostly in your face, neck, and belly.
- **High blood pressure:** Blood pressure numbers that stay high or are hard to control.
- **Skin changes:** Skin getting thin with easy bruising or pink stretch marks.
- **Muscle weakness:** Feeling weak, especially in the upper leg

When It Is an Emergency (Adrenal Crisis)

Sometimes low adrenal function becomes a life-threatening emergency called an **adrenal crisis** (a sudden, severe drop in hormones).

- Severe belly or lower back pain
- Sudden vomiting and diarrhea
- Extreme confusion or dizziness
- Fainting or loss of consciousness

Thyroid Adrenal Connection

- Thyroid and adrenal hormones work together closely to control your body's energy, stress response, and metabolism.
- The Thyroid Adrenal Connection explains that when one of these glands is out of balance, the other often follows.
- The adrenal glands, located above the kidneys, produce hormones such as cortisol, which help the body manage stress, regulate metabolism, and maintain blood pressure. During periods of chronic stress, cortisol levels often remain high. In response, the brain may attempt to lower cortisol production to protect the body — but this same mechanism can also suppress thyroid hormone output, leading to symptoms of hypothyroidism.
- This means that when someone experiences long-term stress, the adrenal glands and thyroid gland can both become imbalanced. Treating only the thyroid without addressing adrenal health can be misleading and may even worsen symptoms over time, sometimes resulting in the need for higher doses of thyroid medication.

How Adrenal Dysfunction Affects Thyroid Function?

The adrenal and thyroid glands communicate through the hypothalamic-pituitary-adrenal-thyroid (HPAT) axis — a complex hormonal feedback system. When stress is prolonged:

- Cortisol levels rise, signaling the body to slow metabolism and conserve energy.
- Conversion of T4 (inactive thyroid hormone) to T3 (active thyroid hormone) decreases.
- Levels of reverse T3 (rT3) may increase, which blocks thyroid receptors and prevents thyroid hormones from working effectively.
- Over time, this leads to what is sometimes called functional hypothyroidism, even when standard lab values appear normal

How They Connect

- T4 to T3 Conversion: High stress and excess cortisol block the body from turning inactive T4 into active T3.
- Receptor Sensitivity: Excess cortisol can make your cells stop responding properly to thyroid hormones.
- Mutual Support: The adrenal glands also need normal levels of thyroid hormone to work the right way.

STRESS CAN LEAD TO THYROID HORMONE RESISTANCE.

The stress response often leads to increased levels of inflammatory molecules, called cytokines. Studies have shown that this inflammation can make your cells less responsive to thyroid hormones. So even though you are on the correct medication dose and your labs are perfect, your body is not able to respond to your medication. So you still lack energy, you still suffer from brain fog, and you just don't feel like yourself.

THYROID HORMONE LEVELS MAY ALTER THE SECRETION AND METABOLISM OF CORTISOL.

In the Thyroid-Adrenal connection, most of the focus is on how cortisol affects the thyroid. But there is evidence that this is a two way street. Studies have shown a correlation between thyroid hormones and how our body makes and metabolizes cortisol. When thyroid hormones are low, the adrenal gland does not secrete as much cortisol and that cortisol is metabolized much slower. So if you have been focusing on treating your adrenal glands with little success, it may be time to investigate your thyroid.

TESTING YOUR THYROID

- TSH
- Free T₄
- Free T₃
- Reverse T₃
- Anti-TPO (antibodies)
- Anti-TG (antibodies)

TESTING YOUR ADRENAL GLANDS

- Serum cortisol, typically only tested once in the morning
- Salivary cortisol tested at 4 time points in one day
- Urinary cortisol and metabolites tested at 3-4 time points

Assessing and treating your thyroid and adrenal glands can be complicated and take time. It is not as simple as taking this supplement and eating that diet.